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Xiao et al.

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(54) **FOOD CONTAINER AND ANIMAL FOOD DISPENSING DEVICE COMPATIBLE WITH FOOD CONTAINER**

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A01K 5/01 (2006.01)
(Continued)

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(Continued)

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See application file for complete search history.

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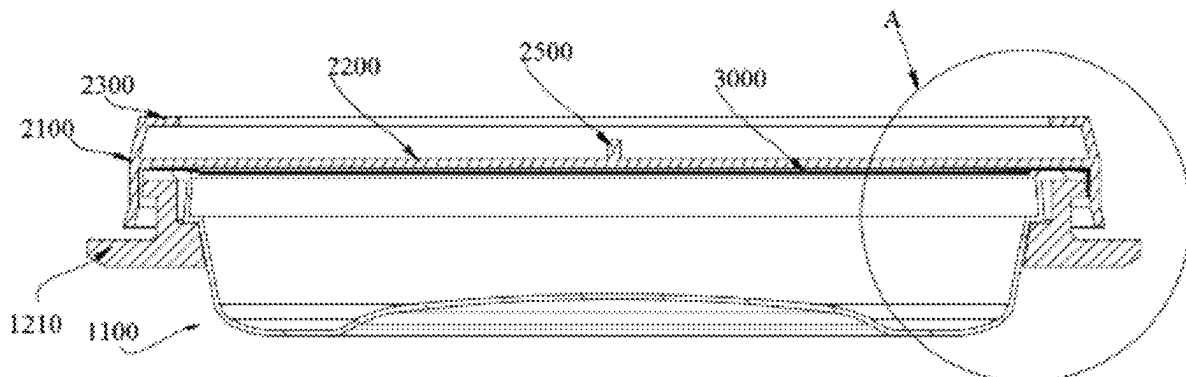
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Primary Examiner — James N Smalley

(57) **ABSTRACT**

A food container and an animal food dispensing device compatible with food container are provided. The food container includes a can body and a cover body. The can body includes a container body including a container bowl portion. The container bowl portion is defined with an accommodation chamber and an opening communicated with the accommodation chamber. The cover body includes a side wall portion, a cover tray portion, and a cover flange portion. The cover tray portion allows the cover body, along an up-down direction of the cover body, to be defined with a recessed space having an opening opened upward and a lower space having an opening opened downward. The cover flange portion, the side wall portion, and the cover tray portion define a groove. The cover body is assembled to the can body, and the container body is snap-fitted with the side wall portion.

8 Claims, 14 Drawing Sheets



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- (52) **U.S. Cl.**
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2543/00537 (2013.01)

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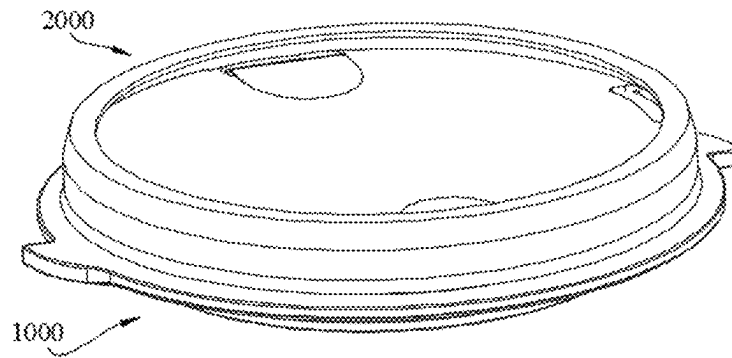


FIG. 1

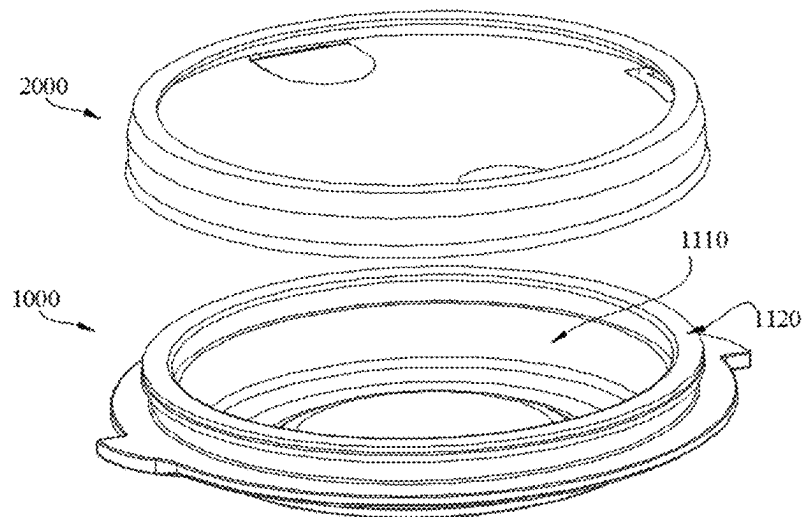


FIG. 2

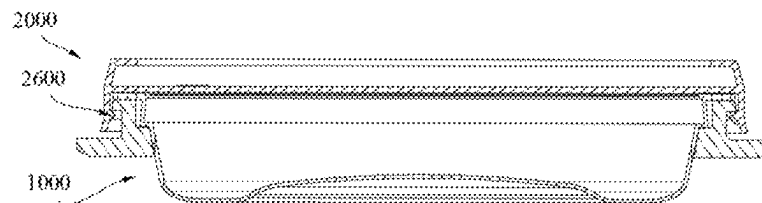


FIG. 3

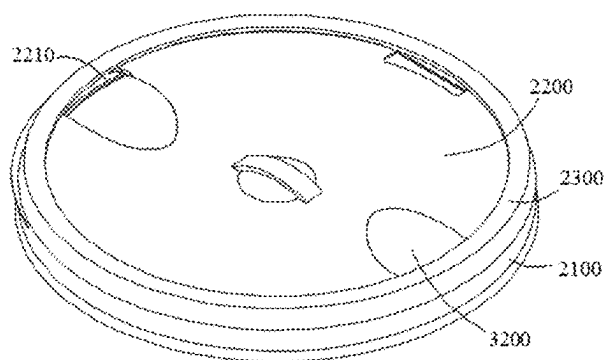


FIG. 4

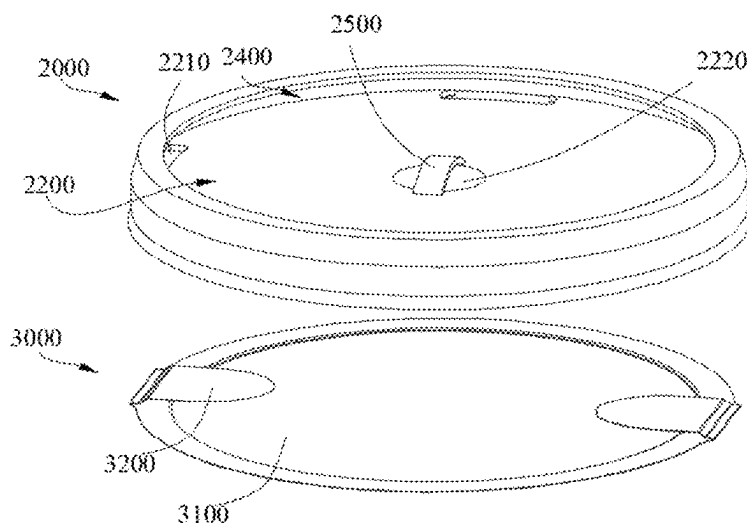


FIG. 5

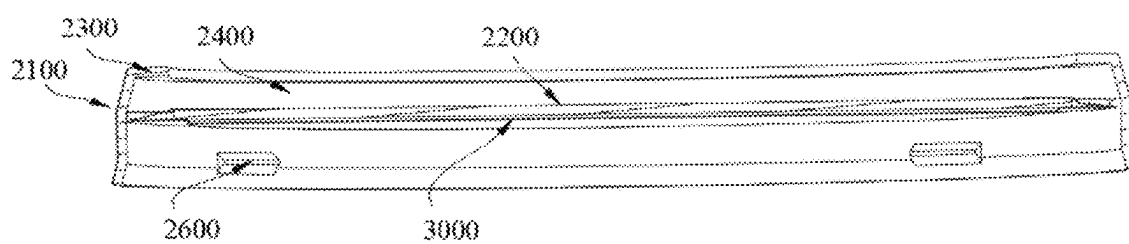


FIG. 6

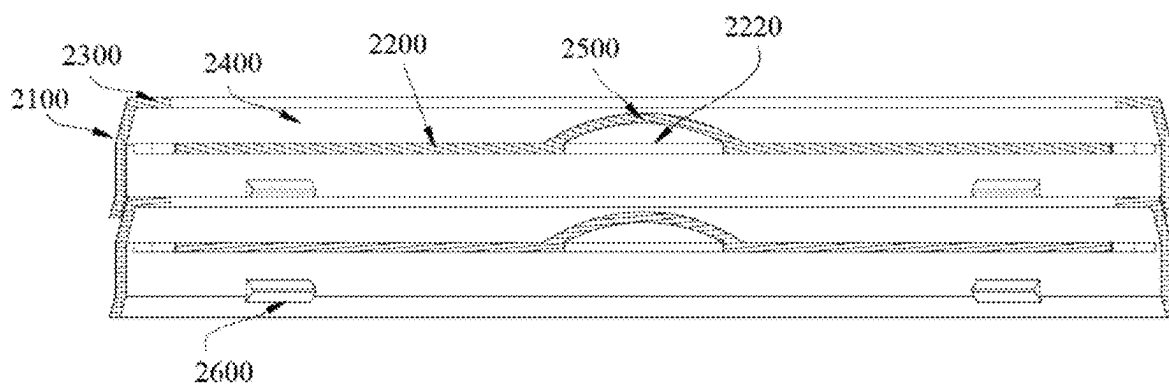


FIG. 7

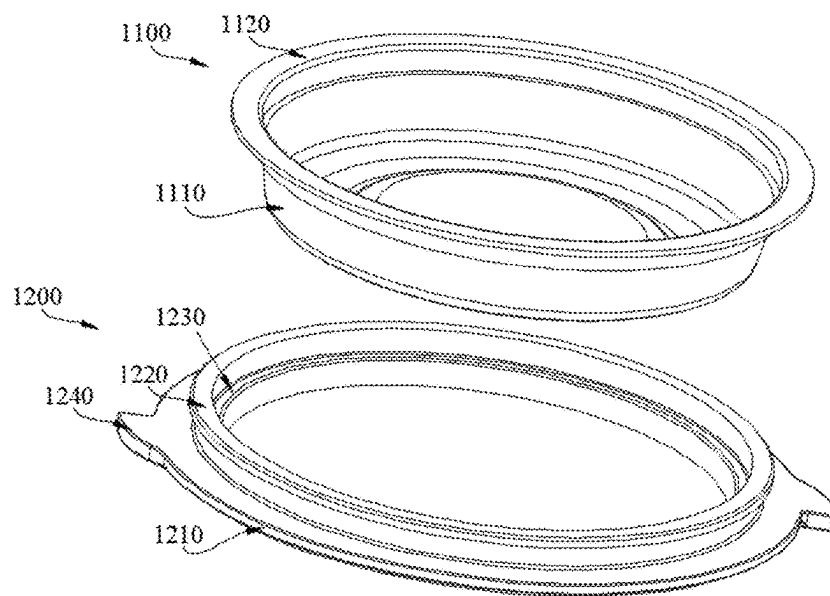


FIG. 8

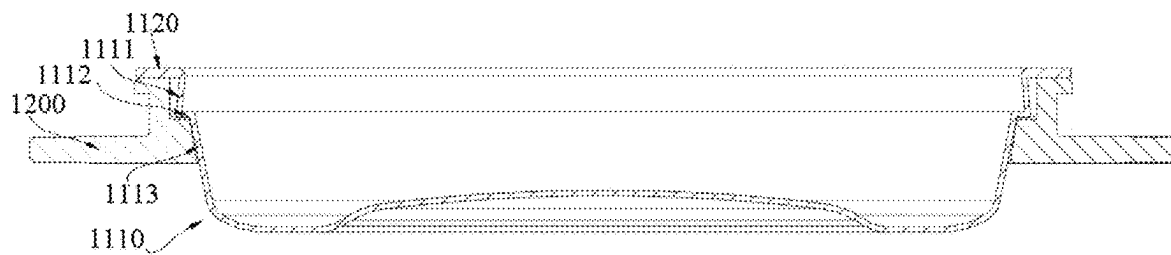


FIG. 9

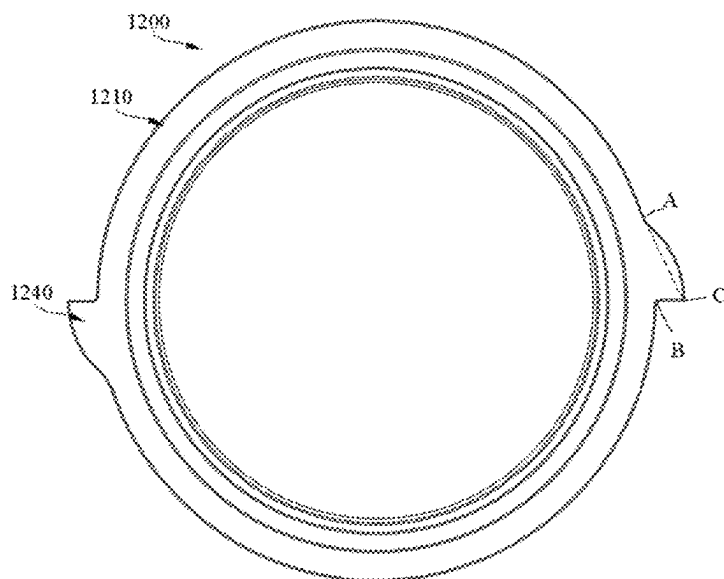


FIG. 10

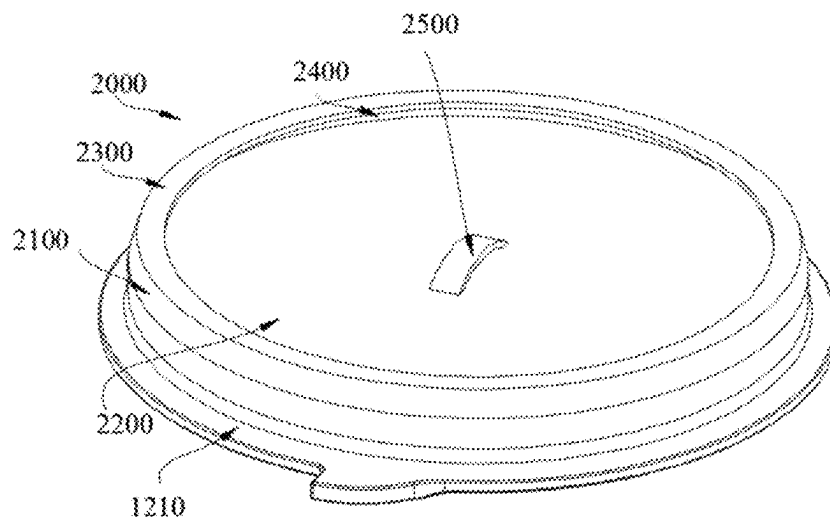


FIG. 11

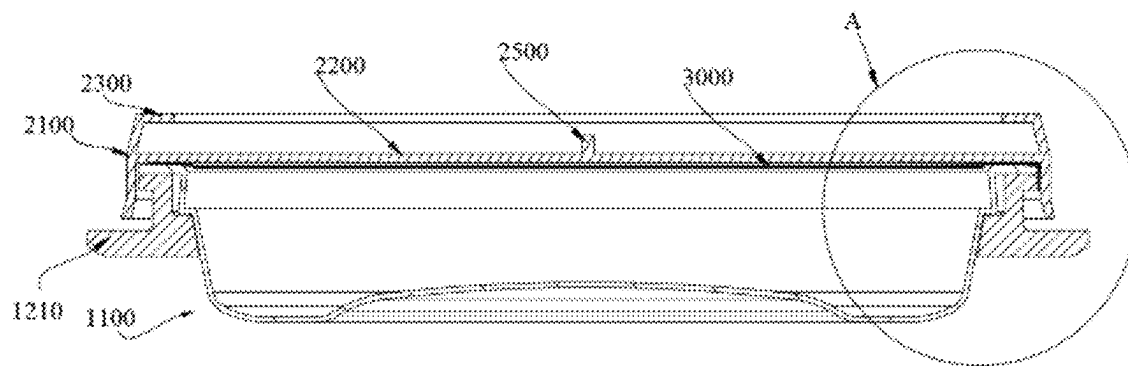


FIG. 12

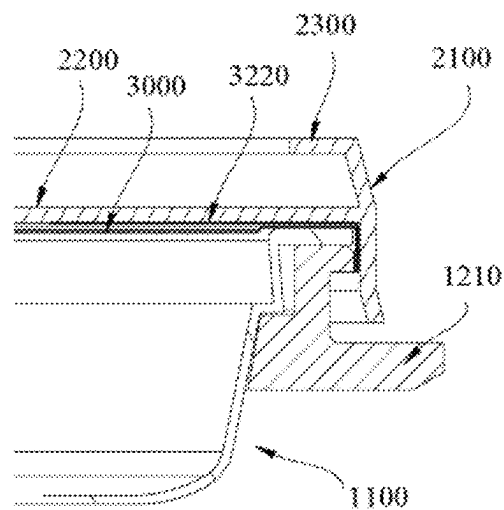


FIG. 13

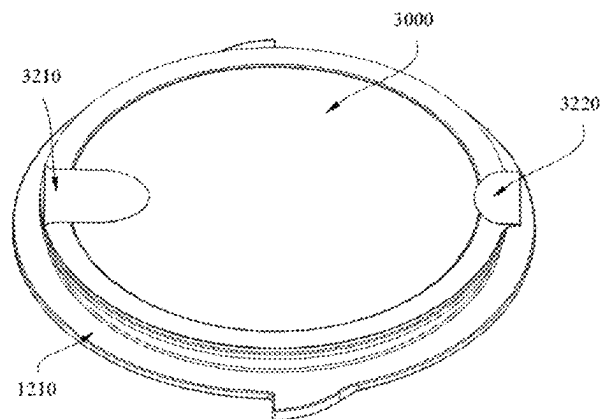


FIG. 14

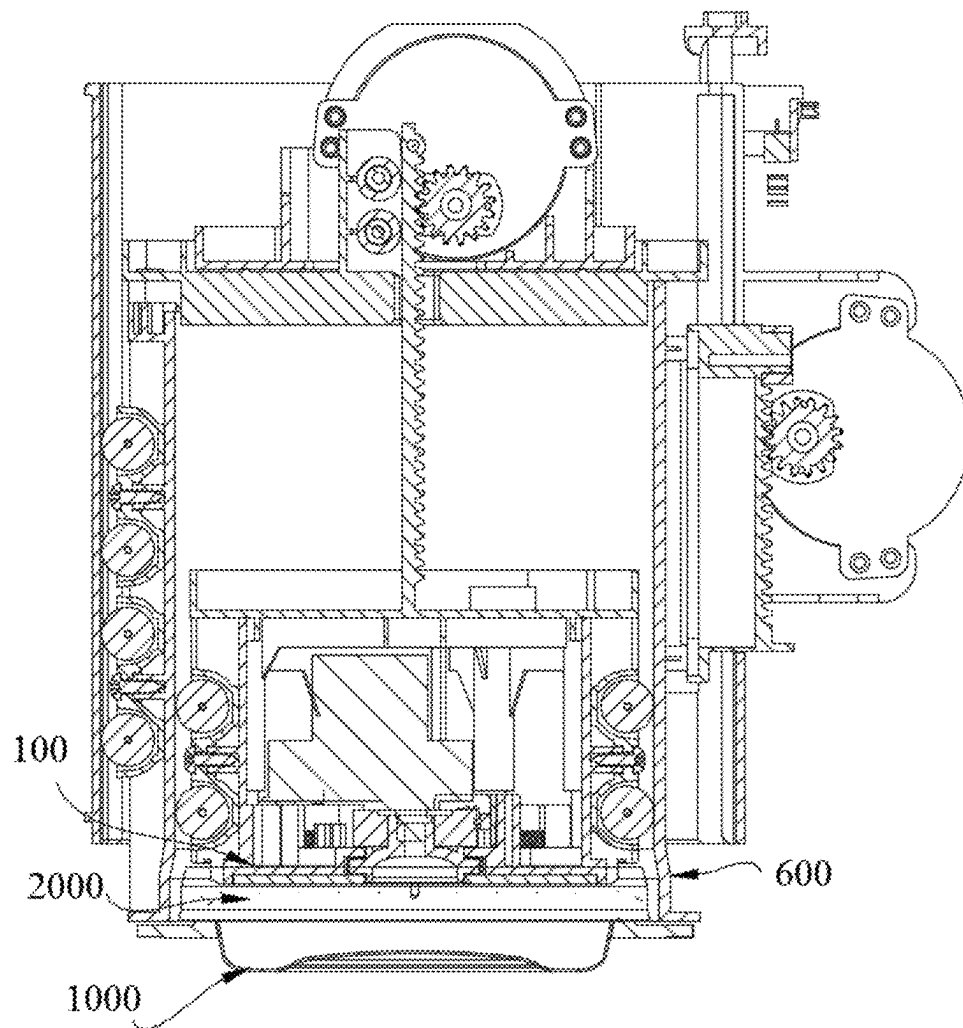


FIG. 15

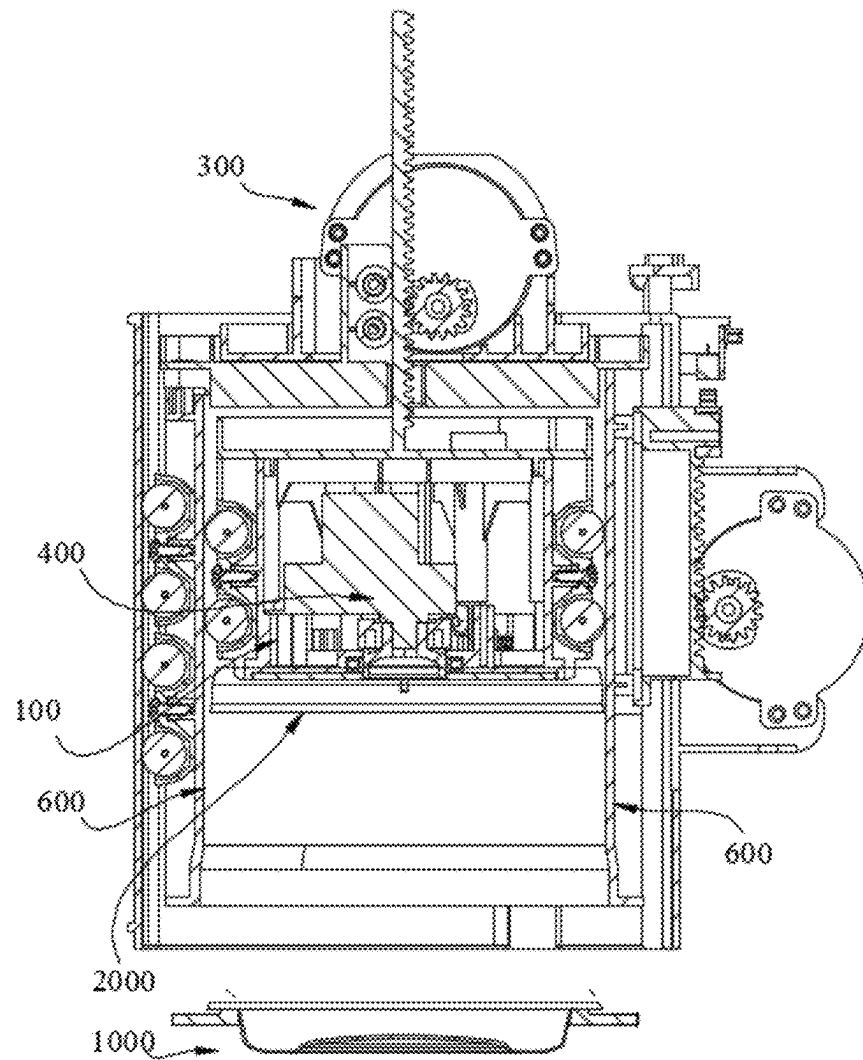


FIG. 16

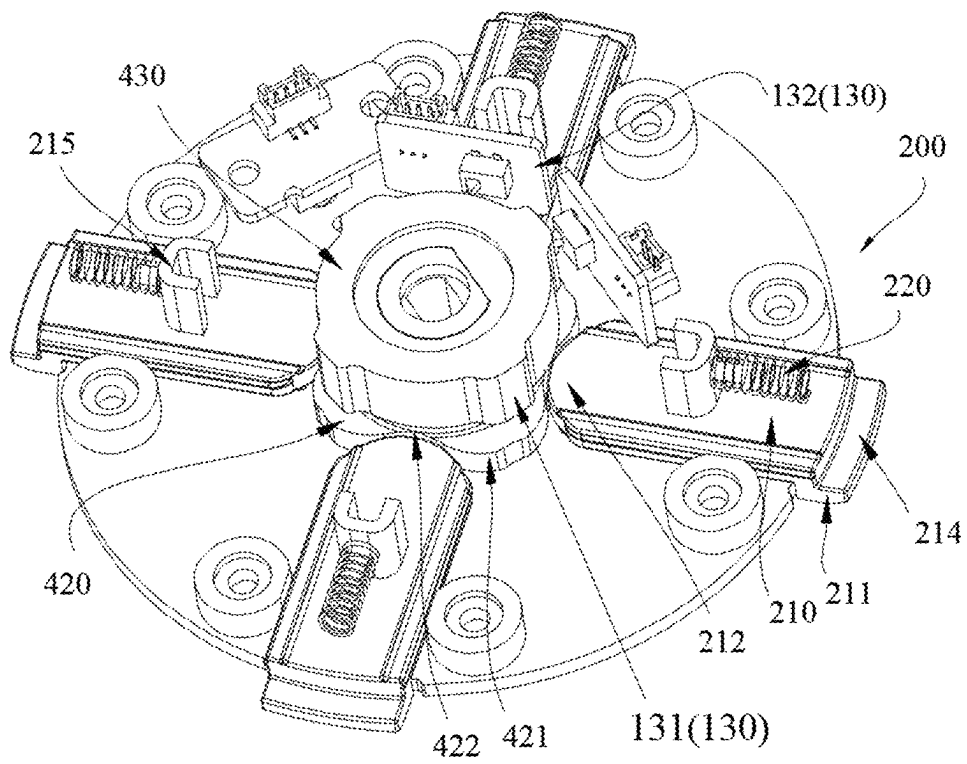


FIG. 17

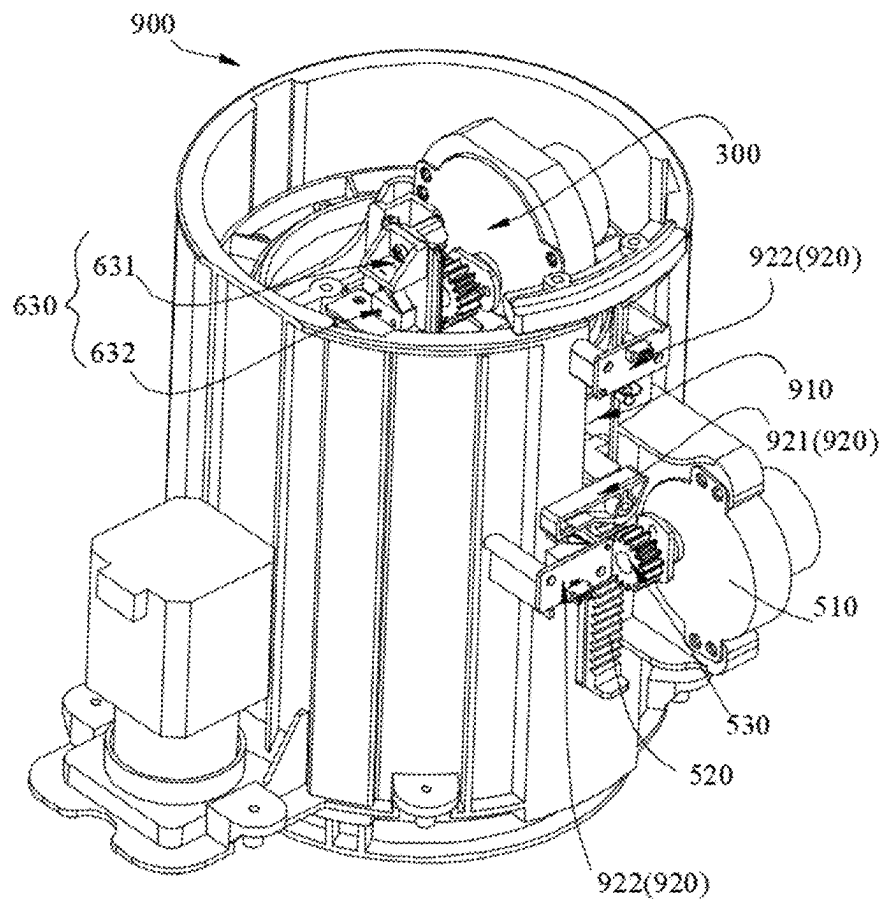


FIG. 18

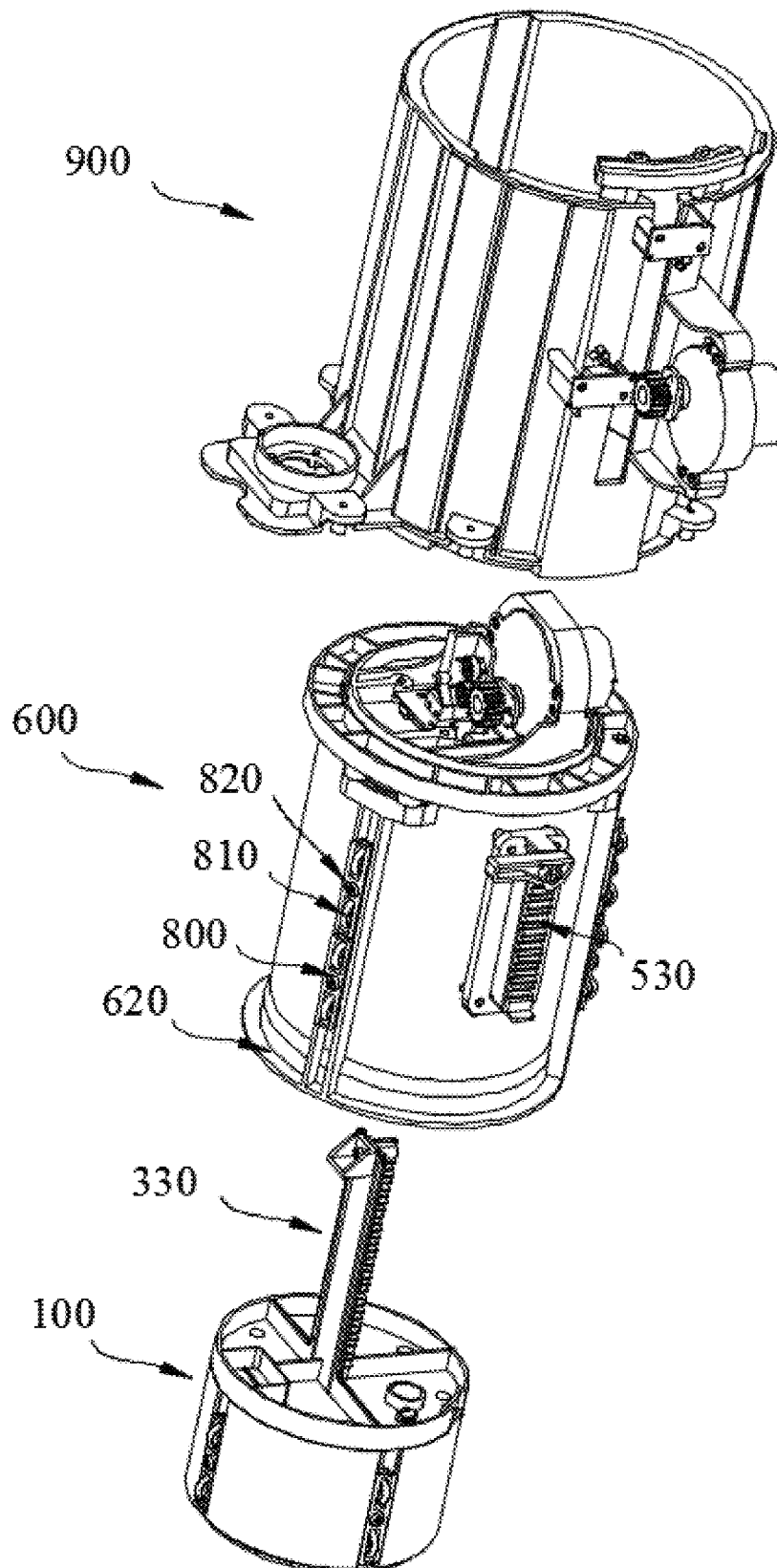


FIG. 19

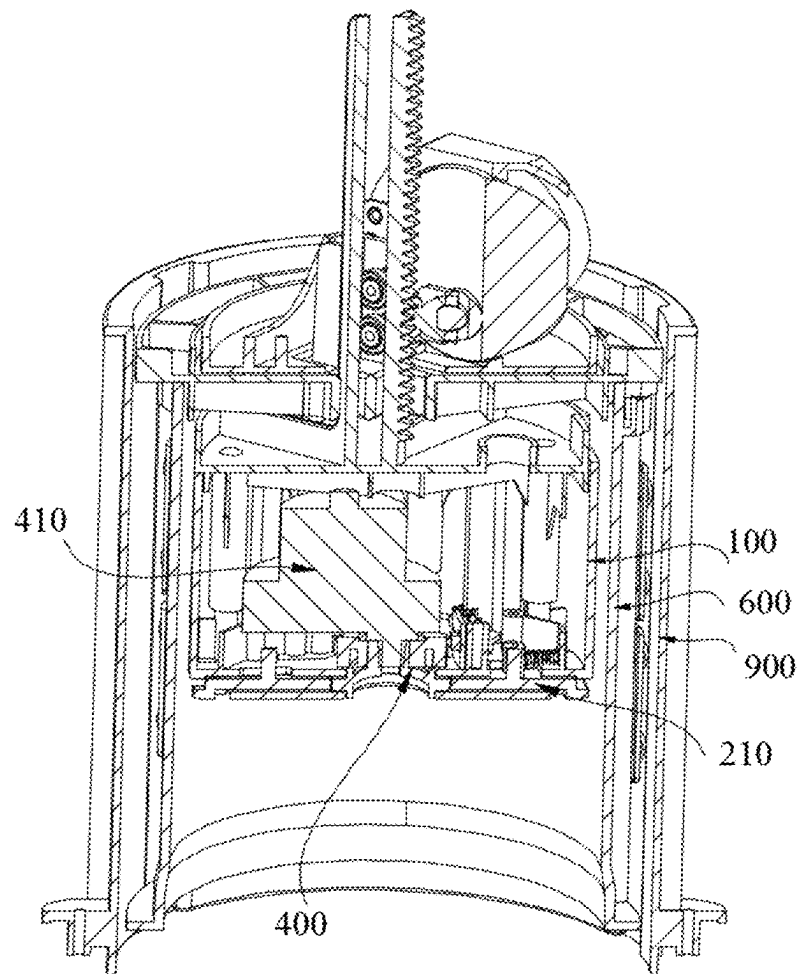


FIG. 20

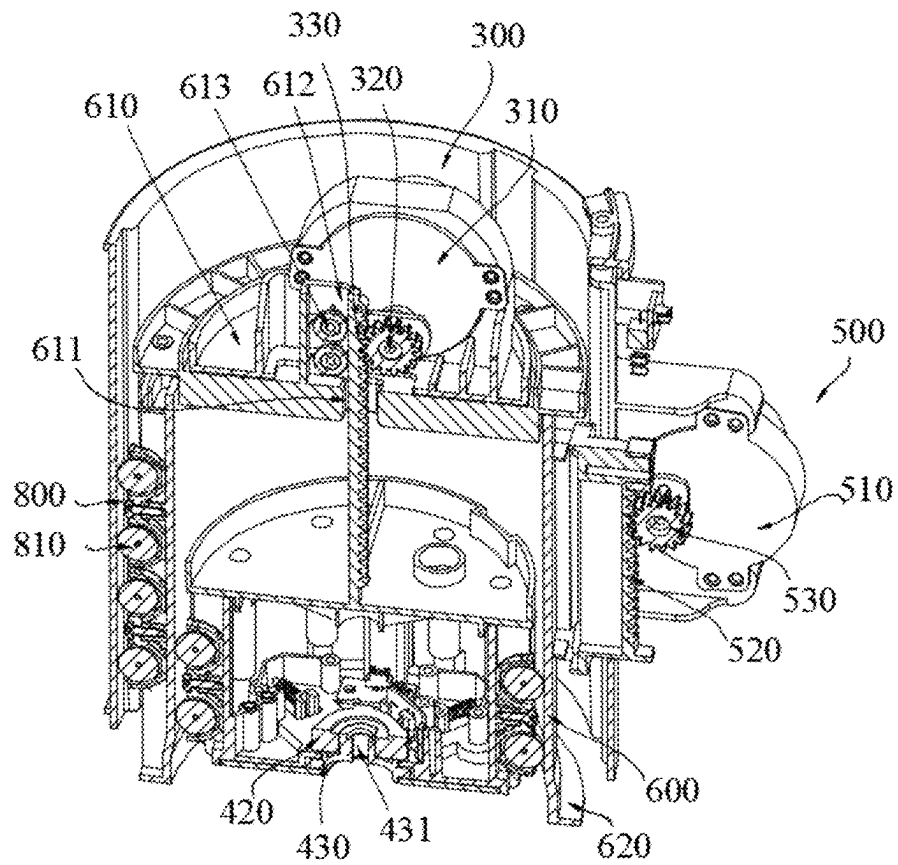


FIG. 21

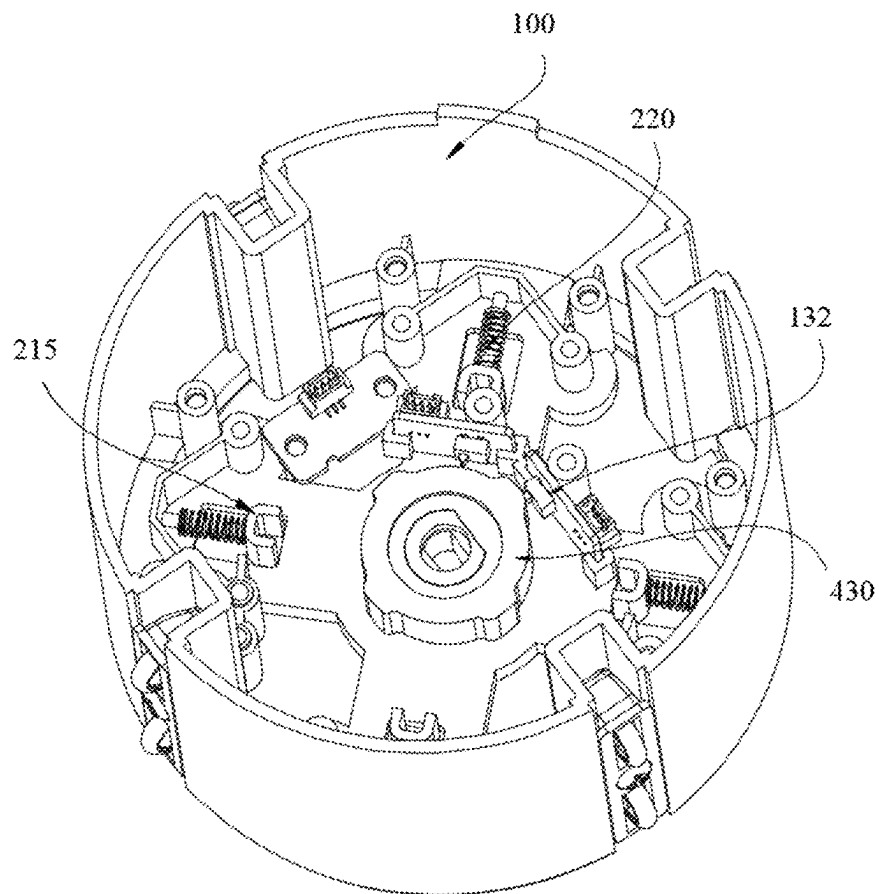


FIG. 22

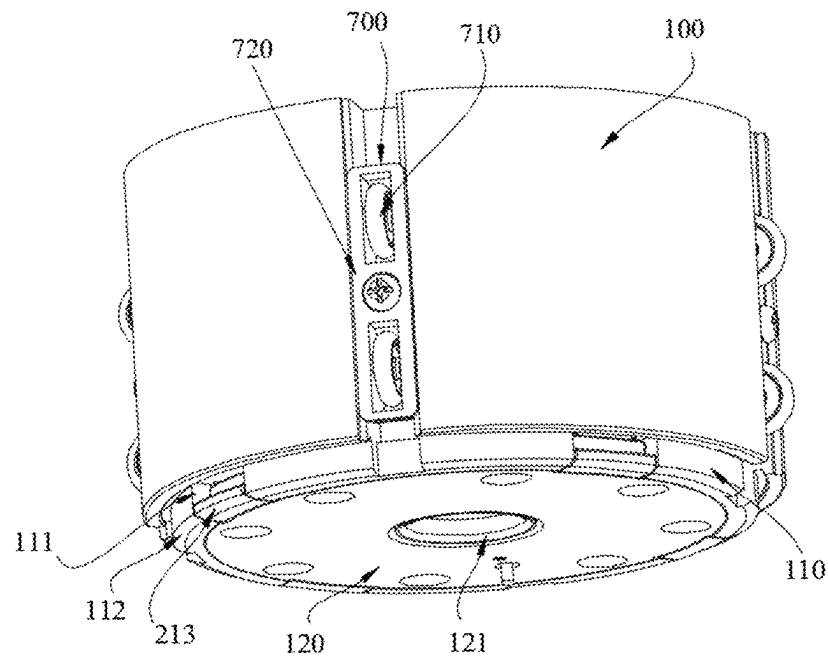


FIG. 23

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FOOD CONTAINER AND ANIMAL FOOD DISPENSING DEVICE COMPATIBLE WITH FOOD CONTAINER

CROSS REFERENCE TO RELATED APPLICATIONS

The present disclosure is a Continuation Application of PCT Application No. PCT/CN2024/080294, filed Mar. 6, 2024, which claims the priority of Chinese Patent Application No. 202420273925.2, filed Feb. 4, 2024 with the National Intellectual Property Administration and entitled “food container and animal food dispensing device compatible with food container”, the entirety of which is hereby incorporated herein by reference for all purposes.

TECHNICAL FIELD

The present disclosure relates to the technical field of food containers for pets, and in particular to a food container and an animal food dispensing device compatible with food container.

BACKGROUND

Pets are becoming indispensable companions in people's lives. Nowadays, the number of pet owners is on the rise. When feeding pets, in addition to cat food, pet canned food and other types of food are also provided. Food containers, such as pet cans, are generally used to hold wet food, such as minced meat, but of course, can also be used for holding dry food. For this type of food container, the cover is typically integrated with the entire container and is opened by means of a pull ring, which is inconvenient and is prone to accidents, and is not suitable for pet feeders that need to automatically open the containers.

SUMMARY

The present disclosure aims to address the technical problems in related art. Therefore, the present disclosure provides a food container suitable for an automatic can-opening mechanism and with a cover that is easy to open.

To realize the above objective, according to a first aspect, the present disclosure provides a food container, including:

a can body, the can body including a container body, the container body comprising a container bowl portion, and the container bowl portion being defined with an accommodation chamber for holding food and an opening opened upward and communicated with the accommodation chamber; and

a cover body, the cover body including a side wall portion, a cover tray portion, and a cover flange portion, the cover tray portion being connected to an inner peripheral wall of the side wall portion, the cover tray portion allowing the cover body, along an up-down direction of the cover body, to be defined with a recessed space having an opening opened upward and a lower space having an opening opened downward, the cover flange portion being arranged in the recessed space, an end of the cover flange portion being connected to the side wall portion, the other end of the cover flange portion extending inward in a direction away from the side wall portion, and the cover flange portion, the side wall portion and the cover tray portion defining a groove; the cover body is assembled to the can body, and the container body is snap-fitted with the side wall portion.

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In some embodiments, the food container further includes a sealing membrane for sealing the accommodation chamber. A portion of the sealing membrane is connected to the cover tray portion, and at least part of the other portion of the sealing membrane is connected to the container bowl portion.

In some embodiments, the sealing membrane includes a sealing membrane body and a connection lug portion extending outward from an outer peripheral edge of the sealing membrane body. The sealing membrane body is connected to an upper surface of the container bowl portion, and the connection lug portion is folded upward and inward to be connected to a lower surface of the cover tray portion.

In some embodiments, the connection lug portion includes a first adhesion section and a second adhesion section connected to the first adhesion section, and the first adhesion section is closer to an outer peripheral edge of the sealing membrane body than the second adhesion section. The connection lug portion is one of two connection lug portions, the first adhesion section of one of the two connection lug portions is connected to the lower surface of the cover tray portion, and the second adhesion section of the other one of the two connection lug portions is connected to the lower surface of the cover tray portion.

In some embodiments, the sealing membrane has a first surface and a second surface facing the first surface, the first surface of the sealing membrane is connected to an upper surface of the container bowl portion, and the second surface of the sealing membrane is connected to a lower surface of the cover tray portion. Or, the cover tray portion is defined with a first through hole, the sealing membrane includes a sealing membrane body and a connection lug portion extending outward from an outer peripheral edge of the sealing membrane body, the sealing membrane body is connected to an upper surface of the container bowl portion, the connection lug portion passes through the first through hole into the recessed space, and the connection lug portion is connected to an upper surface of the cover tray portion.

In some embodiments, the cover body is further provided with a handle portion arranged in the recessed space. The handle portion is connected to the cover tray portion, and an upper surface of the handle portion is lower than the cover flange portion in an axial direction of a central axis of the cover body.

In some embodiments, the cover tray portion is defined with a second through hole, and the handle portion is arranged above the second through hole.

In some embodiments, the second through hole is located at a center of the cover body, within a projection of the food container on a horizontal plane.

In some embodiments, the cover body is further provided with a first flange portion arranged in the lower space, an end of the first flange portion is connected to the side wall portion, and the other end of the first flange portion extends inward in a direction away from the side wall portion. The container body is further provided with a second flange portion, an end of the second flange portion is connected to the container bowl portion, and the other end of the second flange portion extends outward in a direction away from the container body. The cover body is assembled to the can body, and the first flange portion is snap-fitted with the second flange portion.

In some embodiments, the second flange portion forms an upper peripheral edge of the container body. The cover body is assembled to the can body, and an inner edge of the cover flange portion fails to protrude beyond an inner surface of

the second flange portion, within a projection of the food container on a horizontal plane.

In some embodiments, the food container is configured to hold dry food or wet food. And/or, the cover flange portion forms an upper peripheral edge of the cover body. And/or, the groove is an annular groove. And/or, the food container is configured to be compatible with an automatic feeding assembly.

In the technical solution of the present disclosure, the container bowl portion is provided in the can body for holding food, and the cover tray portion of the cover body covers the opening of the container bowl portion. When the cover body is assembled on the can body, the container body is snap-fitted with the side wall portion to facilitate the opening of the cover body. Moreover, the cover flange portion, the side wall portion, and the cover tray portion form the groove, which is convenient for an external animal food dispensing device to be connected to the cover body, to drive the cover body away from the can body. Therefore, the food container provided by the present disclosure can be used with the external animal food dispensing device to automatically open the cover body.

According to a second aspect, the present disclosure provides an animal food dispensing device compatible with food container. The animal food dispensing device is applied to the food container according to any of the foregoing embodiments. The food container further includes a sealing membrane for sealing the accommodation chamber. A portion of the sealing membrane is connected to the cover body, and another portion of the sealing membrane is connected to the can body. The animal food dispensing device includes a press-can apparatus and a pull-cover apparatus. The press-can apparatus is configured to abut against the can body to limit movement of the can body in a vertical direction.

The pull-cover apparatus includes:

- a pull-cover body, the pull-cover body being movably connected with a snap-fit assembly, the snap-fit assembly having a locked state and a released state, and the snap-fit assembly being switchable between the locked state and the released state; in the locked state, the snap-fit assembly being engaged in the groove, and in the released state, the snap-fit assembly being separated from the groove; and
- a first driving assembly, the first driving assembly being in transmission connection with the pull-cover body, to allow the pull-cover body to have a first position and a second position in the vertical direction; at the first position, the snap-fit assembly being configured to engage with the groove, and at the second position, the cover body being separated from the can body and driving the sealing membrane to separate from the can body; and the first driving assembly driving the pull-cover body to move between the first position and the second position.

In some embodiments, the pull-cover apparatus further includes a second driving assembly. The second driving assembly includes a second motor and a first cam. The second motor is connected to the first cam to drive the first cam to rotate, an outer peripheral wall of the first cam is provided with a protruding portion extending in a direction perpendicular to an axial direction of the second motor and a recessed portion. The snap-fit assembly includes an engaging member. The engaging member includes an engaging portion and a pushing portion connected to each other. In the locked state, the first cam drives the protruding portion to abut against the pushing portion, and the pushing portion drives the engaging portion to engage with the groove; and

in the released state, the first cam drives the recessed portion to abut against the pushing portion, and the pushing portion drives the engaging portion to separate from the groove.

In some embodiments, the snap-fit assembly further includes a reset member. The engaging member is further provided with a fixing portion extending upward along an axial direction of a motor shaft of the second motor. An end of the reset member is connected with the pull-cover body, and the other end of the reset member is connected with the fixing portion. The reset member applies an elastic restoring force to the engaging member, to allow the engaging member to switch from the locked state to the released state. And/or, the pull-cover body is in a shape of a sleeve with an internal first accommodation space, the second driving assembly and the snap-fit assembly are arranged in the first accommodation space, a guide rail is provided on a lower wall surface of the first accommodation space, and the engaging member is in sliding fit with the guide rail. And/or, the engaging portion is provided with a first guide inclined surface and a support surface connected with the first guide inclined surface. The first guide inclined surface extends upward from an edge of the engaging member to the support surface; and the first guide inclined surface is configured to cooperate with the cover body during the pull-cover body moving to the first position, to guide the cover body into the support surface.

In some embodiments, the second driving assembly further includes a second cam. The first cam includes a cam shaft. The second cam is sleeved on the cam shaft, and the second motor drives the first cam and the second cam to move synchronously. The animal food dispensing device further includes a first detection assembly. The first detection assembly includes a first signal emitter and a first signal receiver. The first signal emitter is fixedly connected to the second cam, the first signal receiver is fixedly connected to the pull-cover body, and in the locked state or/and the released state, the first signal emitter is provided in correspondence with the first signal receiver.

In some embodiments, the first detection assembly is a mechanical switch, the first signal emitter is a mechanical protrusion, and the first signal receiver is a mechanical button. And/or, the first detection assembly includes two first signal receivers. The number of the first signal emitters is the same as the number of the engaging members and is denoted as N. The first signal emitters and the engaging members are evenly spaced along an axis of the second motor, within a projection on a horizontal plane. The line connecting the axis of the second motor to one of the first signal receivers is L1, and the line connecting the axis of the second motor to the other first signal receiver is L2, where the central angle defined between L1 and L2 is α , and α and N satisfy the equation: $\alpha=180/N$.

In some embodiments, a bottom end of the pull-cover body is provided with a guide portion for cooperating with the cover body. The guide portion is configured to move in an inner peripheral wall of the cover body, to allow the pull-cover body to move into or out of the cover body. The guide portion is defined with a first through hole, and in the locked state, the engaging portion extends out of the first through hole to engage with the groove.

In some embodiments, the guide portion is provided with a second guide inclined surface, the second guide inclined surface is inclined from a bottom of the pull-cover body in a vertical upward direction toward a direction close to the press-can apparatus, and the second guide inclined surface is configured to cooperate with the cover body during the

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pull-cover body moving to the first position, to allow the pull-cover body to move into the cover body.

In some embodiments, the press-can apparatus includes a pressing member and a third driving assembly. The third driving assembly is in transmission connection with the pressing member, to allow the pressing member to have a third position and a fourth position in the vertical direction. At the third position, the pressing member abuts against the first flange portion to limit movement of the can body in the vertical direction; and at the fourth position, the pressing member is separated from the first flange portion. The third driving assembly drives the pressing member to move between the third position and the fourth position.

In some embodiments, the first driving assembly includes a first motor and a first transmission member in transmission connection with the first motor. The first transmission member is fixedly connected with the pull-cover body, and the first motor drives the first transmission member to move in the vertical direction. The pressing member is in a shape of a sleeve with an internal second accommodation space, an end of the pressing member close to the food container is open, and the other end of the pressing member is defined with a second through hole penetrating up and down, the pull-cover body is arranged in the second accommodation space and the first transmission member passes through the second through hole to drive the pull-cover body to move in the vertical direction.

In some embodiments, the animal food dispensing device further includes a fixing member. The pressing member and the pull-cover body are fixedly connected with the fixing member. The fixing member is opened with a second through hole.

In some embodiments, the first transmission member includes a first gear and a first rack. The first motor is mounted on the fixing member. An output shaft of the first motor is in transmission connection with the first gear. The first rack has a first surface and a second surface facing the first surface. The first surface is meshed with the first gear for transmission. The surface of the fixing member facing the first motor is provided with a limiting portion, and the limiting portion is arranged close to the second surface to limit the first rack to move in a direction away from the first gear. The first rack passes through the second through hole to drive the pull-cover body to move vertically.

In some embodiments, the limiting portion includes a third roller, which is cooperated with the first rack to reduce friction.

In some embodiments, the animal food dispensing device further includes a second detection assembly. The second detection assembly includes a second signal emitter and a second signal receiver. The second signal emitter is fixedly connected to the first rack, and the second signal receiver is fixedly connected to the fixing member. The second signal emitter is a mechanical protrusion, and the second signal receiver is a mechanical button. At the first position, the second signal emitter abuts against the second signal receiver. And/or, the bottom of the pressing member is provided with a flanging extending outward along the circumferential direction. The flanging is configured to abut against the support portion of the can body. And/or, the animal food dispensing device further includes a first walking assembly. The first walking assembly includes a first roller, which is assembled to the pull-cover body. The first roller is in rolling cooperation with the pressing member to reduce friction generated when the pull-cover body and the pressing member move relative to each other. And/or, the third position is below the fourth position.

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In some embodiments, the third driving assembly includes a third motor and a second transmission member. The second transmission member includes a second gear and a second rack. An output shaft of the third motor is in transmission connection with the second gear, the pressing member is fixedly connected with the second rack, and the second gear is meshed with the second rack to drive the pressing member to move in the vertical direction.

In some embodiments, the animal food dispensing device further includes a housing in a shape of a sleeve with an internal third accommodation space. The third motor is mounted on the housing, and the press-can apparatus and the pull-cover apparatus are arranged in the third accommodation space. The animal food dispensing device further includes a second walking assembly. The second walking assembly includes a second roller, which is assembled on the housing. The second roller is in rolling operation with the pressing member to reduce friction generated when the pressing member moves relative to the housing.

In some embodiments, the animal food dispensing device further includes a third detection assembly. The third detection assembly includes a third signal emitter and two third signal receivers. The third signal emitter is fixedly connected to the press-can apparatus, and the third signal receivers are fixedly connected to the housing. At the third position, one of the two third signal receivers is provided in correspondence with the third signal receiver; and at the fourth position, the other of the two third signal receivers is provided in correspondence with the third signal receiver.

In some embodiments, the third detection assembly is a photoelectric sensor, the third signal emitter is a photoelectric emitter, and the third signal receiver is a photoelectric receiver; or, the third detection assembly is a magnetic field sensor, the third signal emitter is a magnetic signal emitter, and the third signal receiver is a magnetic signal receiver; or, the third detection assembly is a mechanical switch, the third signal emitter is a mechanical protrusion, and the third signal receiver is a mechanical button.

In the technical solution of the present disclosure, the container bowl portion is provided in the can body for holding food, and the cover tray portion of the cover body covers the opening of the container bowl portion. When the cover body is assembled on the can body, the container body is snap-fitted with the side wall portion to facilitate the opening of the cover body. Moreover, the cover flange portion, the side wall portion, and the cover tray portion form the groove, which is convenient for an external animal food dispensing device to be connected to the cover body, to drive the cover body away from the can body. Therefore, the food container provided by the present disclosure can be used with the external animal food dispensing device to automatically open the cover body.

BRIEF DESCRIPTION OF THE DRAWINGS

In order to explain the technical solutions of the embodiments of the present disclosure more clearly, the following accompanying drawings are briefly described. Obviously, the accompanying drawings are only some embodiments of the present disclosure, for those skilled in the field, other drawings can be obtained based on these drawings without any creative effort.

FIG. 1 is a schematic structural diagram of a food container according to an embodiment of the present disclosure.

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FIG. 2 is an exploded structural diagram of a food container according to an embodiment of the present disclosure.

FIG. 3 is a cross-sectional diagram of a food container according to an embodiment of the present disclosure.

FIG. 4 is a schematic structural diagram of a cover body of a food container according to an embodiment of the present disclosure.

FIG. 5 is an exploded structural diagram of a cover body of a food container according to an embodiment of the present disclosure.

FIG. 6 is a cross-sectional diagram of a cover body of a food container according to an embodiment of the present disclosure.

FIG. 7 is a cross-sectional diagram illustrating stacking of multiple cover bodies of a food container according to an embodiment of the present disclosure.

FIG. 8 is an exploded structural diagram of a can body of a food container according to an embodiment of the present disclosure.

FIG. 9 is a cross-sectional diagram of a can body of a food container according to an embodiment of the present disclosure.

FIG. 10 is a top view of a can body of a food container according to an embodiment of the present disclosure.

FIG. 11 is a schematic structural diagram of a food container according to another embodiment of the present disclosure.

FIG. 12 is a cross-sectional diagram of a food container according to another embodiment of the present disclosure.

FIG. 13 is an enlarged view of portion A in FIG. 12.

FIG. 14 is a schematic structural diagram of a sealing membrane and a can body of a food container according to another embodiment of the present disclosure.

FIG. 15 is a cross-sectional diagram of an animal food dispensing device compatible with food container at a first position according to an embodiment of the present disclosure.

FIG. 16 is a cross-sectional diagram of an animal food dispensing device compatible with food container at a second position according to an embodiment of the present disclosure.

FIG. 17 is schematic structural diagram of a snap-fit assembly of an animal food dispensing device compatible with food container according to an embodiment of the present disclosure.

FIG. 18 is a schematic structural diagram of an animal food dispensing device compatible with food container according to an embodiment of the present disclosure.

FIG. 19 is an exploded structural diagram of an animal food dispensing device compatible with food container according to an embodiment of the present disclosure.

FIG. 20 is a cross-sectional diagram of an animal food dispensing device compatible with food container according to an embodiment of the present disclosure.

FIG. 21 is a cross-sectional diagram of an animal food dispensing device compatible with food container according to another embodiment of the present disclosure.

FIG. 22 is a schematic structural diagram of a pull-cover body of an animal food dispensing device compatible with food container according to an embodiment of the present disclosure from a perspective.

FIG. 23 is a schematic structural diagram of a pull-cover body of an animal food dispensing device compatible with food container according to an embodiment of the present disclosure from another perspective.

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The purpose, functional features, and advantages of the present disclosure will be further illustrated in conjunction with embodiments, with reference to the drawings.

DETAILED DESCRIPTION OF THE EMBODIMENTS

The technical solutions in the embodiments of the present disclosure will be described clearly and completely with reference to the accompanying drawings in the embodiments of the present disclosure. Obviously, the described embodiments are only a part of the embodiments of the present disclosure rather than all of them. Based on the embodiments in the present disclosure, all other embodiments obtained by those skilled in the art without creative work shall fall within the scope of protection of the present disclosure.

It should be noted that all directional indications (such as up, down, left, right, front, back) in the embodiments of the present disclosure are merely configured to explain relative position relationships or motion conditions between the components in a specific attitude (as shown in the drawings). The directional indication changes as the specific attitude changes.

It should be noted that unless otherwise explicitly defined and limited, terms such as “connected” and “fixed” should be understood in a broad sense. For example, “fixed” can mean a rigid connection, a detachable connection, or being integrated into one piece; it may be a mechanical connection or an electrical connection; it may be a direct connection or an indirect connection through an intermediary; it may be an internal connection within two components or an interactive relationship between two components, unless otherwise specifically limited. For those skilled in the art, the specific meaning of the aforementioned terms in the present disclosure can be understood according to the specific circumstances.

Moreover, the terms “first”, “second”, and the like in the present disclosure are merely used for description and cannot be understood as indicating or implying their relative importance or as implicitly indicating the quantity of the technical features indicated. Thus, the feature defined by “first” or “second” may explicitly or implicitly include at least one such feature. Furthermore, the term “and/or” throughout the description includes three concurrent scenarios. For example, “A and/or B” includes scenario A alone, scenario B alone, or both A and B together. Additionally, the technical solutions of various embodiments may be combined with each other, but must be based on that the combined technical solutions can be implemented by those skilled in the art. When the combination of the technical solutions is contradictory or impossible to realize, it shall be considered that such combination does not exist and is not within the scope of protection of the present disclosure.

The present disclosure provides a food container. In related art, the cover of the food container is inconvenient to open and is not suitable for an animal food dispensing device that automatically opens the food container.

To solve the above technical problems, in the technical solution of the present disclosure, the container bowl portion is provided in the can body for holding food, and the cover tray portion of the cover body covers the opening of the container bowl portion. When the cover body is assembled on the can body, the container body is snap-fitted with the side wall portion to facilitate the opening of the cover body. Moreover, the cover flange portion, the side wall portion, and the cover tray portion form the groove, which is

convenient for an external animal food dispensing device to be connected to the cover body, to drive the cover body away from the can body. Therefore, the food container provided by the present disclosure can be used with the external animal food dispensing device to automatically open the cover body.

The above technical solutions will be described in detail below with reference to the accompanying drawings.

In the embodiments, as shown in FIG. 1 to FIG. 6, the food container includes a can body **1000** and a cover body **2000**. The can body **1000** includes a container body **1100**. The container body **1100** includes a container bowl portion **1110**. The container bowl portion **1110** is defined with an accommodation chamber for holding food and an opening opened upward and communicated with the accommodation chamber. It should be noted that food is placed in the container bowl portion **1110**, and the cover body **2000** is fastened on the can body **1000**. The can body **1000** is detachably connected with the cover body **2000**. When it is necessary to feed a pet with food from the container bowl portion **1110**, the cover body **2000** is driven to separate from the can body **1000**, exposing the opening. In this way, the pet can consume the food placed in the container bowl portion **1110** through the opening. In a specific implementation process, the accommodation chamber of the container bowl portion **1110** can be configured to hold dry food or wet food.

Referring to FIG. 4 to FIG. 6, the cover body **2000** includes a side wall portion **2100**, a cover tray portion **2200**, and a cover flange portion **2300**. The cover tray portion **2200** is connected to an inner peripheral wall of the side wall portion **2100**. The cover tray portion **2200** allows the cover body **2000**, along an up-down direction of the cover body **2000**, to be defined with a recessed space having an opening opened upward and a lower space having an opening opened downward. The cover flange portion **2300** is arranged in the recessed space. One end of the cover flange portion **2300** is connected to the side wall portion **2100**, and the other end of the cover flange portion **2300** extends inward in a direction away from the side wall portion **2100**. A groove **2400** is defined by the cover flange portion **2300**, the side wall portion **2100**, and the cover tray portion **2200**. When the cover body **2000** is assembled to the can body **1000**, the container body **1100** is snap-fitted with the side wall portion **2100**.

In a specific implementation process, the shape of the cover tray portion **2200** is adapted to fit the opening of the accommodation chamber. The side wall portion **2100** is arranged along the side wall edge of the cover tray portion **2200** and surrounds the cover tray portion **2200**. The upper and lower ends of the side wall portion **2100** are respectively located on two sides of the cover tray portion **2200**, to form the recessed space on one side of the cover tray portion **2200** and the lower space on the other side. An opening is defined in the upper part of the recessed space, which is convenient for the animal food dispensing device to connect with the cover body **2000** and drive the cover body **2000** to separate from the can body **1000**. An opening is defined in the lower part of the lower space, and the top of the can body **1000** is accommodated in the lower space through the opening. The edge of the cover flange portion **2300** is connected to the top edge of the side wall portion **2100** and extends in a direction close to the axis of the cover body **2000**, such that the groove **2400** is formed by the cover flange portion **2300**, the side wall portion **2100** located on the side of the recessed space, and the cover tray portion **2200**. The cover flange portion **2300** forms the upper peripheral edge of the cover body **2000**. The animal food dispensing device can extend into the

groove **2400** and is engaged with the cover body **2000** to pull the cover body **2000**. This allows the cover body **2000** to be separated from the can body **1000**, thereby opening the food container. It should be noted that, in order for the animal food dispensing device to effectively engage with the cover body **2000**, in a specific implementation process, the cover flange portion **2300** extends in the direction close to the axis of the cover body **2000**, and the cover flange portion **2300** is horizontally arranged. Specifically, the bottom wall of the cover flange portion **2300** facing the groove **2400** is horizontally arranged, enhancing the stability of the engagement.

The side wall portion **2100** is configured to be snap-fitted with the container body **1100**, thereby realizing the detachable connection between the cover body **2000** and the can body **1000**. The snap-fit between the side wall portion **2100** and the container body **1100** may be implemented by a snap-fit structure. For example, one of the side wall portion **2100** and the container body **1100** is provided with a protrusion, and the other one is defined with a groove snap-fitted with the protrusion; or, one of the side wall portion **2100** and the container body **1100** is connected with a snap fastener, and the other one is connected with another snap fastener mated with the aforementioned snap fastener, which is not limited herein.

As shown in FIG. 3 and FIG. 6, the cover body **2000** is further provided with a first flange portion **2600**. The first flange portion **2600** is arranged in the lower space. One end of the first flange portion **2600** is connected to the side wall portion **2100**, and the other end of the first flange portion **2600** extends inward in a direction away from the side wall portion **2100**. The container body **1100** is further provided with a second flange portion **1120**. One end of the second flange portion **1120** is connected to the container bowl portion **1110**, and the other end of the second flange portion **1120** extends outward in a direction away from the container body **1100**. When the cover body **2000** is assembled to the can body **1000**, the first flange portion **2600** is snap-fitted with the second flange portion **1120**.

In the embodiments, the cover body **2000** is snap-fitted with the can body **1000** by the first flange portion **2600** and the second flange portion **1120**. In some embodiments, the first flange portion **2600** is a protrusion that protrudes from the side wall portion **2100** and is located on the side of the side wall portion **2100** facing the lower space. The second flange portion **1120** is a protrusion protruding from the container bowl portion **1110** or a protruding strip structure surrounding the peripheral side wall. The first flange portion **2600** is snap-fitted with the second flange portion **1120**. In another embodiment, one end of the first flange portion **2600** is a fixed end, and the other end of the first flange portion **2600** is a free end. The fixed end is connected to the side wall portion **2100**, and the free end extends away from the surface of the side wall portion **2100**, such that the first flange portion **2600** protrudes from the surface of the side wall portion **2100**. The second flange portion **1120** is arranged with reference to the first flange portion **2600**, to ensure smooth snap-fit between the first flange portion **2600** and the second flange portion **1120**.

In a specific implementation process, to enhance the strength and effectiveness of the snap-fit, the first flange portion **2600** is integrally formed with the side wall portion **2100**, and the second flange portion **1120** is integrally formed with the container bowl portion **1110**. Moreover, to enhance the strength of the snap-fit, multiple first flange portions **2600** are arranged at intervals along the circumferential direction of the cover body **2000**, and the second

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flange portion 1120 extends along the axis of the container bowl portion 1110 and surrounds the container bowl portion 1110.

Further, based on the aforementioned, in the embodiments, the second flange portion 1120 forms the upper peripheral edge of the container body 1100. When the cover body 2000 is assembled to the can body 1000, the inner edge of the cover flange portion 2300 does not protrude beyond the inner surface of the second flange portion 1120, within a projection of the food container on a horizontal plane. According to the embodiments, when the width of the engagement part of the groove 2400 is constant, compared to other solutions (e.g., the inner edge of the cover flange portion 2300 protrudes beyond the inner surface of the second flange portion 1120), the stacking of multiple cover bodies 10 is more stable, thus being beneficial for transportation and assembly in the factory. The stacking of multiple cover bodies 10 is shown in FIG. 7.

It can be understood that bacteria and moisture in the air can affect the food in the container bowl portion 1110, making it prone to spoilage. Therefore, it is necessary to seal the containment chamber of the container bowl portion 1110 to extend the shelf life of the food. In some embodiments, as shown in FIG. 5 and FIG. 6, the food container further includes a sealing membrane 3000 for sealing the accommodation chamber, to improve the sealing effect of the accommodation chamber. In addition, when opening the food container, it is necessary to pull the cover body 2000 and the sealing membrane 3000 in sequence, which is inconvenient, especially not suitable for the animal food dispensing device that automatically opens the food container. Further, in the embodiments, a portion of the sealing membrane 3000 is connected to the cover tray portion 2200, and at least part of the other portion of the sealing membrane 3000 is connected to the container bowl portion 1110. In this way, the sealing membrane 3000 is connected to the cover tray portion 2200 while sealing the accommodation chamber. Thus, when the cover body 2000 is driven to separate from the can body 1000, the sealing membrane 3000 is also separated from the container bowl portion 1110 to expose the food, eliminating the need to open the sealing membrane 3000 again. Moreover, the sealing membrane 3000 may be connected by means of adhesive attachment or heat fusion, which depends on an actual requirement and the material of the sealing membrane 3000. In a specific implementation process, to ensure that the sealing membrane 3000 has a certain strength to avoid damage, an aluminum membrane is used for the sealing membrane 3000.

Specifically, in some embodiments, the sealing membrane 3000 has a first surface and a second surface facing the first surface. The first surface of the sealing membrane 3000 is connected to an upper surface of the container bowl portion 1110, and the second surface of the sealing membrane 3000 is connected to a lower surface of the cover tray portion 2200. In some embodiments, the first surface of the sealing membrane 3000 is connected to the upper surface of the container bowl portion 1110 by means of adhesive attachment. It can be understood that, a ring of adhesive is applied on the part of the first surface of the sealing membrane 3000 close to its edge along the perimeter, and it is bonded to the upper surface of the container bowl portion 1110. An adhesive may be applied at any position on the second surface to bond with the lower surface of the cover tray portion 2200. The strength of the adhesive bonding between the second surface and the cover tray portion 2200 ensures that when the cover body 2000 is driven to move away, the sealing

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membrane 3000 can release its adhesion to the container bowl portion 1110 and move synchronously with the cover body 2000.

In another embodiment, referring to FIG. 5, the cover tray portion 2200 is defined with a first through hole 2210, and the sealing membrane 3000 includes a sealing membrane body 3100 and a connection lug portion 3200 extending outward from an outer peripheral edge of the sealing membrane body 3100. The sealing membrane body 3100 is connected to the upper surface of the container bowl portion 1110. The connection lug portion 3200 passes through the first through hole 2210 into the recessed space, and the connection lug portion 3200 is connected with the upper surface of the cover tray portion 2200.

In a specific implementation process, to improve the strength of the sealing membrane 3000, the connection lug portion 3200 is integrally formed with the sealing membrane body 3100. It can be understood that the part of the sealing membrane body 3100 close to its edge is connected with the top surface of the container bowl portion 1110. To allow the cover body 2000 to smoothly drive the sealing membrane 3000 to move, the connection lug portion 3200 is connected with the edge of the sealing membrane body 3100, to allow the force of the cover body 2000 to directly act on the edge of the sealing membrane body 3100, further releasing the connection between the sealing membrane body 3100 and the container bowl portion 1110. The cover tray portion 2200 is provided with the first through hole 2210 close to its edge, the end of the connection lug portion 3200 away from the sealing membrane body 3100 enters the recessed space through the first through hole 2210, and one surface of the connection lug portion 3200 is connected to the surface of the cover tray portion 2200 facing the recessed space. By using the connection lug portion 3200, the pulling force of the cover body 2000 can be directly applied to the connection of the sealing membrane body 3100 and the container bowl portion 1110, making it easier to release the connection between the two. Further, to ensure balanced and effective separation of the sealing membrane body 3100 from the container bowl portion 1110, in a specific implementation process, the sealing membrane body 3100 is provided with at least two sets of connection lug portions 3200, which are symmetrically arranged.

Referring to FIG. 11 to FIG. 14, in another embodiment of the food container, the sealing membrane 3000 includes a sealing membrane body 3100 and a connection lug portion 3200 extending outward from an outer peripheral edge of the sealing membrane body 3100. The sealing membrane body 3100 is connected with the upper surface of the container bowl portion 1110, and the connection lug portion 3200 is folded upward and inward to connect with the lower surface of the cover tray portion 2200. It can be understood that the part of the sealing membrane body 3100 close to its edge is connected with the top surface of the container bowl portion 1110. To allow the cover body 2000 to smoothly drive the sealing membrane 3000 to move, the connection lug portion 3200 is connected with the edge of the sealing membrane body 3100, and the connection lug portion 3200 is folded upward and inward to connect with the cover body 2000. This allows the force of the cover body 2000 to be transferred to the edge of the sealing membrane body 3100 via the connection lug portion 3200, thereby further releasing the connection between the sealing membrane body 3100 and the container bowl portion 1110.

In the embodiments, as shown in FIG. 14, the connection lug portion 3200 includes a first adhesion section 3210 and a second adhesion section 3220 connected to each other. The

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first adhesion section **3210** is closer to the outer peripheral edge of the sealing membrane body **3100** than the second adhesion section **3220**. The sealing membrane **3000** includes two connection lug portions **3200**, the first adhesion section **3210** of one of the two connection lug portions **3200** is connected to the lower surface of the cover tray portion **2200**, and the second adhesion section **3220** of the other one of the two connection lug portions **3200** is connected to the lower surface of the cover tray portion **2200**. It will be appreciated that the upward movement of the cover body **2000** causes the two connection lug portions **3200** of the sealing membrane **3000** to be subjected to different forces, which is advantageous for maintaining the structural stability of the food container.

To ensure effective separation of the sealing membrane body **3100** from the container bowl portion **1110**, in a specific implementation process, the sealing membrane **3000** includes multiple connection lug portions **3200**, which are symmetrically arranged around the central axis of the sealing membrane **3000**.

As shown in FIG. 13, when the sealing membrane **3000** is assembled to the can body **1000** and the cover body **2000**, the outer edge of the sealing membrane **3000** (such as the part where the connection lug portion **3200** is folded when the second adhesion section **3220** of the connection lug portion **3200** is connected with the lower surface of the cover tray portion **2200**) protrudes beyond the outer peripheral surface of the container body **1100**, within a projection of the food container on a horizontal plane. The can body **1000** covers and accommodates the outer part of the sealing membrane **3000** protruding beyond the outer peripheral surface of the container body **1100**.

Certainly, based on the foregoing embodiments, it can be understood that the connection between the sealing membrane **3000** and the container bowl portion **1110**, as well as between the sealing membrane **3000** and the cover tray portion **2200** may be realized by means of vacuum suction, as long as the accommodation chamber of the container bowl portion **1110** maintains a certain level of airtightness.

In some embodiments, the cover body **2000** is further provided with a handle portion **2500**. The handle portion **2500** is arranged in the recessed space, and is connected to the cover tray portion **2200**. The upper surface of the handle portion **2500** is lower than the cover flange portion **2300** in the axial direction of the central axis of the cover body **2000**.

The handle portion **2500** is configured for hand-holding, which is convenient for lifting the cover body **2000**. In order to improve the strength of the handle portion **2500**, in a specific implementation process, the handle portion **2500** is integrally formed with the cover tray portion **2200**. In practice, the handle portion **2500** is an arc-shaped strip structure, with two ends connected to the cover tray portion **2200**. In addition, an extension direction from either end toward the center is an arc, allowing the middle part of the handle portion **2500** to protrude upward and be higher than the two ends. This creates a certain space between the handle portion **2500** and the cover tray portion **2200**, to facilitate the user's hand to reach into the space between the handle portion **2500** and the cover tray portion **2200**, thereby lifting the handle portion **2500**.

Further, the cover tray portion **2200** is defined with a second through hole **2220**, and the handle portion **2500** is arranged above the second through hole **2220**. Specifically, the handle portion **2500** is arranged across the second through hole **2220** to increase the space below the handle portion **2500**, facilitating the user's hand to pull the cover body **2000**.

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In a specific implementation process, within a projection of the food container on a horizontal plane, the second through hole **2220** is located at the center of the cover body **2000**, and the handle portion **2500** is arranged across the second through hole **2220**, with two ends being at an equal distance from the second through hole **2220**. In this way, the center of the handle portion **2500** is located at the center of the cover body **2000**. By using the handle portion **2500** to pull the cover body **2000**, the force can be evenly transferred to the edge of the cover body **2000**, which allows the cover body **2000** to smoothly separate from the can body **1000**, enhancing the stability of opening the food container.

Referring to FIG. 8 to FIG. 10, a support member **1200** is fixedly sleeved on the outer peripheral wall of the container bowl portion **1110**, and one end of a third flange portion **1210** is connected with the container bowl portion **1110**. The support member **1200** may be connected with the container bowl portion **1110** by means of adhesive fixation; or, the support member **1200** may be locked and fixed on the outer peripheral wall of the container bowl portion **1110** by a connection member such as a screw or a pin. In the embodiments, the support member is bonded to the container body; that is, adhesive dots are applied between the two to achieve a tight and secure bonding connection. In a specific implementation process, the support member **1200** is configured to abut against the animal food dispensing device for opening the food container. Specifically, the other end of the support member **1200** protrudes beyond the outer peripheral wall of the container body **1100**, such that the part between two ends of the support member **1200** protrudes beyond the outer peripheral wall of the container body **1100**, which facilitates the animal food dispensing device to abut against the support member **1200**. In a specific implementation process, the outer edge of the third flange portion **1210** protrudes beyond the outer peripheral surface of the cover body **2000** within a projection of the food container on a horizontal plane, preventing interference of the container bowl portion **1110**. This allows the animal food dispensing device, when pulling to open the cover body **2000**, to limit the movement of the can body **1000** by abutting against the support member **1200**, facilitating the animal food dispensing device to drive the cover body **2000** away from the can body **1000** so as to open the food container for pet feeding.

In some embodiments, the support member **1200** is not integrally formed with the container body **1100**. It will be appreciated that the support member **1200** is fixedly connected to the container body **1100**, such that the animal food dispensing device abuts against the support member **1200** to limit the movement of the container body **1100**. In a specific implementation process, on the one hand, the production of the container body **1100** is independent from the production of the support member **1200**, the container body **1100** has its own mold, and the support member **1200** also has its own mold, which simplifies the design and manufacturing of the molds. On the other hand, the container bowl portion **1110** of the container body **1100** is used for holding food, which means that the material selection for the container body **1100** is quite strict to meet the hygiene requirements for pet food; while the support member **1200** does not come into contact with the food, so there is a wider variety of material options available for it. This allows for more implementation flexibility in the industrialization process.

The cover body **2000** is snap-fitted with the container body **1100**. In some embodiments, the container body **1100** is further provided with the second flange portion **1120**. One end of the second flange portion **1120** is connected to the container bowl portion **1110**, and the other end of the second

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flange portion 1120 extends outward in a direction away from the container body 1100. The outer edge of the second flange portion 1120 protrudes beyond the outer peripheral surface of the container bowl portion 1110 within a projection of the food container on a horizontal plane. The cover body 2000 is further provided with the first flange portion 2600. One end of the first flange portion 2600 is connected to the side wall portion 2100, and the other end of the first flange portion 2600 extends inward in a direction away from the side wall portion 2100. When the cover body 2000 is assembled to the can body 1000, the first flange portion 2600 is snap-fitted with the second flange portion 1120.

In a specific implementation process, the second flange portion 1120 is arranged around the peripheral side wall of the container body 1100. In the embodiments, the second flange portion 1120 conforms to the contour of the outer peripheral wall of the container body 1100. Specifically, the second flange portion 1120 is a ring-shaped structure, which is arranged around the edge of the container body 1100 close to the opening. The inner edge of the second flange portion 1120, which is the ring-shaped structure, is connected with the outer peripheral side wall of the container bowl portion 1110, and the outer edge of the second flange portion 1120 protrudes beyond the outer peripheral surface of the container bowl portion 1110, such that the second flange portion extends horizontally away from the container bowl portion 1110. It can be understood that the second flange portion 1120 can serve as a flanging structure on the upper end of the container bowl portion 1110. Further, the second flange portion 1120 forms the upper peripheral edge of the container body 1100. In practice, the second flange portion 1120 is integrally formed with the container bowl portion 1110, thereby enhancing the overall strength and improving the connection strength between the cover body 2000 and the can body 1000.

In another embodiment, the support member 1200 is further provided with a fourth flange portion 1220. The fourth flange portion 1220 forms the upper peripheral edge of the support member 1200. The outer edge of the fourth flange portion 1220 protrudes beyond the outer peripheral surface of the container bowl portion 1110 within a projection of the food container on a horizontal plane. The cover body 2000 is further provided with a first flange portion 2600. One end of the first flange portion 2600 is connected to the side wall portion 2100, and the other end of the first flange portion 2600 extends inward in a direction away from the side wall portion 2100. When the cover body 2000 is assembled to the can body 1000, the fourth flange portion 1220 is snap-fitted with the first flange portion 2600.

Referring to FIG. 8, the fourth flange portion 1220 and the third flange portion 1210 are respectively located at the upper and lower ends of the support member 1200. The fourth flange portion 1220 forms the upper peripheral edge of the support member 1200, and the third flange portion 1210 forms the lower peripheral edge of the support member 1200. The inner end of the fourth flange portion 1220 is close to the container bowl portion 1110, and the other end extends away from the container bowl portion 1110. This allows the outer edge of the fourth flange portion 1220 to protrude beyond the outer peripheral surface of the container bowl portion 1110, which facilitates the snap-fit between the first flange portion 2600 and the fourth flange portion 1220, and prevents interference from the container bowl portion 1110.

In the embodiments, the container body 1100 is further provided with a second flange portion 1120. One end of the second flange portion 1120 is connected to the container

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bowl portion 1110, and the other end of the second flange portion 1120 extends outward in a direction away from the container body 1100. The upper surface of the fourth flange portion 1220 is connected to the lower surface of the second flange portion 1120. The outer edge of the fourth flange portion 1220 protrudes beyond the outer peripheral surface of the second flange portion 1120 within a projection of the food container on a horizontal plane.

Specifically, it should be noted that one end of the second flange portion 1120 is the side wall of the second flange portion 1120 close to the accommodation chamber, which is connected to the outer peripheral surface of the container bowl portion 1110 and extends circumferentially around the container bowl portion 1110. The other end of the second flange portion 1120 extends in a direction away from the container bowl portion 1110 such that the other end of the second flange portion 1120 extends out of the container bowl portion 1110. In practice, it can be understood that the second flange portion 1120 can serve as a flanging structure on the upper end of the container bowl portion 1110, and the second flange portion 1120 is integrally formed with the container bowl portion 1110, thereby enhancing the overall strength and improving the connection strength between the cover body 2000 and the can body 1000. Further, the outer edge of the fourth flange portion 1220 protrudes beyond the outer peripheral surface of the second flange portion 1120, facilitating the snap-fit with the cover body 2000 and providing protection to the second flange portion 1120.

Referring to FIG. 9, in a specific implementation process, a second section 1112 extends in a horizontal direction, and the end portion of the second section 1112 close to the central axis of the container bowl portion 1110 forms a second peripheral wall. A first peripheral wall faces the second peripheral wall. Specifically, one end of the first section 1111 is connected with the first peripheral wall and extends upward to the opening of the accommodation chamber. One end of a third section 1113 is connected with the second peripheral wall and extends downward to the bottom wall of the container bowl portion 1110. In this way, the first section 1111, the second section 1112, and the third section 1113 form the side wall of the container bowl portion 1110, and a step structure is formed at the second section 1112. This structure cooperates with a step portion 1230 of the support member, to allow the support member, namely the step portion 1230 and the parts extending upward and downward from the step portion 1230, to connect with the side wall of the container bowl portion 1110, thereby enhancing the connection strength between the support member 1200 and the container bowl portion 1110.

In practice, the food container is placed inside an animal food dispensing device to allow its cover body 2000 to be automatically opened. It can be understood that the animal food dispensing device is defined with a space or a recess for placing the food container. The animal food dispensing device needs to cooperate with the cover body 2000 and the can body 1000, to allow the cover body 2000 to separate from the can body 1000. There is a certain positioning requirement for the placement of the food container to ensure it is placed in the corresponding position each time for proper cooperation with the animal food dispensing device. In the embodiments, the positioning requirement of the food container is achieved by providing a positioning portion 1240 on the support member 1200.

Specifically, in the embodiments, referring to FIG. 8, the support member 1200 is further provided with a positioning portion 1240 extending outward in a direction away from the container body 1100. The positioning portion 1240 is con-

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nected to an outer side of part of the third flange portion 1210. Define points A and B as where the positioning portion 1240 is connected with the third flange portion 1210, and point C as the outer edge point of the positioning portion 1240. Then, the angle ACB is an acute angle, the length of the segment AC is greater than the length of the segment BC, and the segment AC is an arc segment. In a specific implementation process, to enhance the positioning effect, the support member 1200 is provided with two sets of positioning portions 1240, which are arranged in central symmetry about the center of the support member 1200. Specifically, there is a connecting line L between the mid-point of the point A and the point B and the central point of the container body 1100. The point C of one set of positioning portion 1240 and the point C of the other set of positioning portion 1240 are located on two sides of the connecting line L. This arrangement helps prevent the food container from being inverted and ensures that the food container can be quickly positioned and accurately placed.

Referring to FIG. 15 to FIG. 23, the present disclosure further provides an animal food dispensing device compatible with food container. The animal food dispensing device is applied to the food container described above. The animal food dispensing device includes a press-can apparatus and a pull-cover apparatus. The press-can apparatus is configured to abut against the can body 1000, to limit vertical movement of the can body 1000. The press-can apparatus is located above the food container and abuts against the edge of the can body 1000. Specifically, the press-can apparatus may be a sleeve structure, which uniformly abuts against the top surface of the outer side wall of the can body 1000, thereby limiting the upward movement of the can body 1000.

The pull-cover apparatus includes a pull-cover body 100 and a first driving assembly 300. The pull-cover body 100 is snap-fitted in the groove 2400 of the cover body 2000. The first driving assembly 300 drives the pull-cover body 100 to move in the vertical direction. It should be noted that, in the embodiments, the cover body 2000 of the food container is snap-fitted with the can body 1000, and through the snap-fit between the pull-cover body 100 and the groove 2400 of the cover body 2000, the animal food dispensing device pulls the cover body 2000 away from the can body 1000 to open the food container. Specifically, the first driving assembly 300 drives the pull-cover body 100 to vertically pull the cover body 2000, giving the cover body 2000 a tendency to move away from the can body 1000 in the vertical direction. Meanwhile, the press-can apparatus abuts against the can body 1000 to limit a vertical displacement of the can body 1000. In this way, the pull-cover apparatus can pull the cover body 2000 away from the can body 1000 to open the food container.

In a specific implementation process, with reference to FIG. 13, the pull-cover body 100 is movably connected to a snap-fit assembly 200. The snap-fit assembly 200 has a locked state and a released state, and it is switchable between the locked state and the released state. In the locked state, the snap-fit assembly 200 is engaged in the groove 2400; and in the released state, the snap-fit assembly 200 is separated from the groove 2400. Specifically, the snap-fit assembly 200 includes an engaging member 210, a second driving assembly 400, and a reset member 220. The second driving assembly 400 is drivingly connected with the engaging member 210, and drives one end of the engaging member 210 to extend out of the pull-cover body 100 to be engaged in the groove 2400 of the cover body 2000. The reset member 220 is connected with the engaging member

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210 and drives the engaging member 210 to move in the opposite direction, allowing the engaging member 210 to separate from the cover body 2000. The second driving assembly 400 drives one end of the engaging member 210 to extend out of the pull-cover body 100, such that the engaging member 210 is engaged with the cover body 2000; and the first driving assembly 300 cooperates with the reset member 220, to allow the engaging member 210 to move away from the cover body 2000, such that the engaging member 210 is separated from the cover body 2000. This allows the snap-fit assembly 200 to switch between the locked state and the released state, thereby smoothly opening the food container. It can be understood that the reset member 220 has a certain elastic force. When the second driving assembly 400 drives the engaging member 210 to extend out, the reset member 220 is compressed, which allows it to have a reverse driving force. Therefore, when the engaging member 210 needs to separate from the cover body 2000, the reset member 220 can utilize the reverse driving force to drive the engaging member 210 to move reversely. In practice, the reset member 220 may be a spring, rubber, silicone, or the like, which is not limited herein. One end of the reset member 220 is connected to the engaging member 210, and the other end of the reset member 220 is connected to the pull-cover body 100.

The transmission connection between the first driving assembly 300 and the pull-cover body 100 allows the pull-cover body 100 to have a first position and a second position in the vertical direction. At the first position, the snap-fit assembly 200 is configured to be engaged in the groove 2400; and at the second position, the cover body 2000 is separated from the can body 1000 and drives the sealing membrane 3000 to separate from the can body 1000. The first driving assembly 300 drives the pull-cover body 100 to move between the first position and the second position. Specifically, when it is necessary to open the food container, the first driving assembly 300 drives the pull-cover body 100 to reach the first position, and the bottom of the pull-cover body 100 enters the recessed space. The snap-fit assembly 200 is switched from the released state to the locked state and extends into the groove 2400 of the cover body 2000, achieving the snap-fit. The first driving assembly 300 drives the pull-cover body 100 to move toward the second position such that the cover body 2000 moves away from the can body 1000 to separate from the can body 1000. Meanwhile, the sealing membrane 3000 is connected with the cover body 2000, so as the cover body 2000 moves, it also drives the sealing membrane 3000 to separate from the can body 1000, thereby opening the food container. In this way, the food in the container bowl portion 1110 is exposed for pets to eat. It can be understood that, the first driving assembly 300 may also drive the pull-cover body 100 to descend and carry the cover body 2000 to cover the can body 1000, to protect the unfinished food inside the can body 1000 or facilitate the disposal of the entire food container. Specifically, in the process of the first driving assembly 300 driving the pull-cover body 100 and the cover body 2000 to descend to the first position, the snap-fit assembly 200 remains in the locked state to prevent the cover body 2000 from falling off. After reaching the first position, the snap-fit assembly 200 is switched from the locked state to the released state, which facilitates the pull-cover body 100 to separate from the cover body 2000.

In some embodiments, as shown in FIG. 15 to FIG. 23, the animal food dispensing device compatible with food container is applied to a can. The can includes a cover body 2000 and a can body 1000 snap-fitted with the cover body

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2000. The animal food dispensing device includes a press-can apparatus and a pull-cover apparatus. The press-can apparatus is configured to abut against the can body 1000 to limit vertical movement of the can body 1000. The technical solution is applicable to the can whose cover body 2000 and can body 1000 are snap-fitted with each other. It can be understood that the cover body 2000 and the can body 1000 may move away from each other to open the can; or, one of the two is fixed while the other one moves away to open the can. In the embodiments, the press-can apparatus is configured to abut against the can body 1000 to limit the vertical displacement of the can body 1000, and the pull-cover apparatus is configured to cooperate with the cover body 2000, to allow the cover body 2000 to move vertically and separate from the can body 1000, thereby opening the can.

In the embodiments, the pull-cover apparatus includes a pull-cover body 100 and a first driving assembly 300. Specifically, the pull-cover body 100 is movably connected with a snap-fit assembly 200. The snap-fit assembly 200 has a locked state and a released state, and it is switchable between the locked state and the released state. In the locked state, the snap-fit assembly 200 is engaged with the cover body 2000; and in the released state, the snap-fit assembly 200 is separated from the cover body 2000. The first driving assembly 300 is in transmission connection with the pull-cover body 100 and allows the pull-cover body 100 to have a first position and a second position in the vertical direction. The first position is located below the second position. At the first position, the snap-fit assembly 200 is configured to be engaged in the groove 2400 of the cover body 2000; and at the second position, the cover body 2000 is separated from the can body 1000. The first driving assembly 300 drives the pull-cover body 100 to move between the first position and the second position.

The first driving assembly 300 may be a motor-driven lifting mechanism or a cylinder-driven lifting mechanism, which is not limited herein. In a specific implementation process, referring to FIG. 15 and FIG. 16, the first driving assembly 300 drives the pull-cover body 100 to move in the vertical direction, and allows the pull-cover body 100 to have the first position and the second position within its movement stroke. There is a certain distance between the first position and the second position to ensure the cover body 2000 and the pull-cover body 100 to be completely separated from the can body 1000. It will be appreciated that after the cover body 2000 and the pull-cover body 100 are separated from the can body 1000, the can body 1000 may be moved to a feeding position of the animal food dispensing device, making it available for pet feeding. The cover body 2000 is defined with a groove, and the snap-fit assembly 200 is engaged with the cover body 2000 through the groove, such that when the first driving assembly 300 drives the pull-cover body 100 to move vertically, it pulls the cover body 2000 to move synchronously. At the same time, the press-can apparatus limits the vertical movement of the can body 1000. In this way, the pull-cover body 100 can pull the cover body 2000 away from the can body 1000 to open the can.

In the embodiments, as shown in FIG. 14 or FIG. 19, the can further includes a sealing membrane 3000 for sealing the accommodation chamber. A portion of the sealing membrane 3000 is connected to the cover body 2000, and at least part of the other portion of the sealing membrane 3000 is connected to the can body 1000. In this way, when the animal food dispensing device opens the cover body 2000, the sealing membrane 3000 is separated from the can body 1000 along with the cover body 2000. Specifically, the

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press-can apparatus is configured to abut against the top of a support portion of the can body 1000 to limit the vertical movement of the can body 1000. The pull-cover apparatus cooperates with the cover body 2000 and moves vertically to pull the cover body 2000 away from the can body 1000. Meanwhile, the sealing membrane 3000 moves synchronously with the cover body 2000 to separate from the can body 1000, to expose the food inside the accommodation chamber for pets to consume, achieving the effect of automatically opening the food container. In other embodiments, the can body and the cover body can be used directly to seal the food. In the embodiments, the support portion of the can body 1000 is the third flange portion 1210 of the support member 1200.

In the embodiments, the can is placed in the animal food dispensing device, and it is driven to move under the pull-cover apparatus. The movement of the can may be implemented by a turntable driving the can to rotate, or by manual movement, etc., which is not limited herein. In this case, the pull-cover body 100 is located at a position away from the can, which may be used as an initial position of the pull-cover body 100. When the animal food dispensing device controls to open the can, the first driving assembly 300 drives the pull-cover body 100 to move downward from the initial position to the first position. This movement is taken as a first stroke of the pull-cover apparatus. It can be understood that, in order for the snap-fit assembly 200 to be smoothly engaged into the groove of the cover body 2000, the snap-fit assembly 200 remains in the released state until it reaches the first position. When the pull-cover body 100 reaches the first position, the snap-fit assembly 200 is switched from the released state to the locked state and is engaged with the cover body 2000. After the snap-fit assembly 200 is engaged with the cover body 2000, the first driving assembly 300 drives the pull-cover body 100 to rise. At the same time, the snap-fit assembly 200 pulls the cover body 2000 to rise synchronously. Furthermore, the press-can apparatus limits the movement of the can body 1000, to allow cover body 2000 to separate from the can body 1000, thereby realizing the opening of the can. The movement of the pull-cover body 100 and the cover body 2000 from the first position to the second position is taken as a second stroke of the pull-cover apparatus. It can be understood that, during the second stroke, when the pull-cover body 100 is at the second position, the snap-fit assembly 200 is in the locked state to prevent the cover body 2000 from falling off. The can body 1000 can be transferred to the feeding position of the animal food dispensing device for the pet to consume.

After the pet has finished eating, the can body 1000 moves from the feeding position to below the pull-cover apparatus. The first driving assembly 300 drives the pull-cover body 100 and the cover body 2000 to descend from the second position to the first position. This movement is taken as a third stroke of the pull-cover apparatus. Understandably, during the third stroke, to prevent the cover body 2000 from falling off, the snap-fit assembly 200 remains in the locked state. After the pull-cover body 100 reaches the first position, the cover body 2000 is engaged with the can body 1000. The snap-fit assembly 200 is switched from the locked state to the released state, thereby separating from the cover body 2000. Then, the first driving assembly 300 drives the pull-cover body 100 and the snap assembly 200 to rise to the initial position or the second position. This movement is taken as a fourth stroke of the pull-cover apparatus. It will be appreciated that, in some embodiments, there may be a situation where all the food inside the can body 1000 is consumed; and in this case, the animal food dispensing

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device may drive the can to a disposal position for discarding. In another embodiment, there may be a situation where there is still food left in the can body **1000**; and in this case, the animal food dispensing device may drive the can to a refrigerated storage position to ensure the quality of the food.

In other embodiments, in the process of the pull-cover body **100** moving to the first position, the snap-fit assembly **200** moves downward with the cover body **2000** to abut against the can body **1000**, and the pull-cover body **100** continues to move downward until it reaches the first position. In this process, the pull-cover body **100** can realize the snap-fit between the cover body **2000** and the can body **1000**.

In some embodiments, referring to FIG. **20** and FIG. **21**, the pull-cover apparatus further includes a second driving assembly **400**. The second driving assembly **400** includes a second motor **410** and a first cam **420**. The second motor **410** is connected to the first cam **420**, to drive the first cam **420** to rotate. The outer peripheral wall of the first cam **420** is provided with a protruding portion **421** extending in a direction perpendicular to the axial direction of the second motor **410** and a recessed portion **422**. The snap-fit assembly **200** includes an engaging member **210**, which includes an engaging portion **211** and a pushing portion **212** connected to each other. Specifically, the engaging member **210** extends along the radial direction of the first cam **420**. The pushing portion **212** is arranged at one end of the engaging member **210** close to the first cam **420** for abutting against the protruding portion **421** and the recessed portion **422**, and the engaging portion **211** is arranged at the other end of the engaging member **210** away from the first cam **420** for engagement with the cover body **2000**. In the locked state, the first cam **420** drives the protruding portion **421** to abut against the pushing portion **212**, and the pushing portion **212** drives the engaging portion **211** to be engaged in the groove **2400** of the cover body **2000**. In the released state, the first cam **420** drives the recessed portion **422** to abut against the pushing portion **212**, and the pushing portion **212** drives the engaging portion **211** to be separated from the cover body **2000**.

In a specific implementation process, with reference to FIG. **17** and FIG. **22**, an output shaft of the second motor **410** is connected to the first cam **420**, and drives the first cam **420** to rotate. During the rotation of the first cam **420**, the protruding portion **421** and the recessed portion **422** of the first cam **420** alternately abut against the pushing portion **212** of the engaging member **210**, such that the snap-fit assembly **200** is switched between the locked state and the released state. In the embodiments, multiple engaging members **210** are arranged at intervals along the circumferential direction of the first cam **420** to ensure that the cover body **2000** is smoothly pulled after the engaging members **210** are engaged with the cover body **2000**. Correspondingly, the numbers of the protruding portions **421** and the recessed portions **422** are the same as or more than the number of the engaging members **210** to ensure that each engaging member **210** abuts against one protruding portion **421** or one recessed portion **422** during the rotation of the cam. In the embodiments, the engaging member **210** is engaged in the groove of the cover body **2000**. In the embodiments, as shown in FIG. **16**, in the locked state, the snap-fit assembly **200** is engaged in the groove **2400**; and in the released state, the snap-fit assembly **200** is separated from the groove **2400**. It can be understood that the number of the grooves defined in the cover body **2000** is the same as or greater than the number of the engaging members **210**, or the groove of the

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cover body **2000** is an integral one, ensuring that when in the locked state, each engaging member **210** is engaged within the corresponding groove.

In a specific implementation process, the axis of the second cam **430** is coincident with the axis of the pull-cover body **100**. The extension direction of the engaging member **210** is consistent with the radial direction of the second cam **430**, and the engaging member **210** is movable along the radial direction. Further, the pull-cover body **100** is in the shape of a sleeve having an internal first accommodation space, and the second driving assembly **400** and the snap-fit assembly **200** are arranged in the first accommodation space. This allows for the efficient use of space and helps to reduce the overall volume of the animal food dispensing device. A lower wall surface of the first accommodation space is provided with a guide rail, and the engaging member **210** is slidably fitted with the guide rail. Specifically, in some embodiments, the lower wall surface of the pull-cover body **100** is opened with a sliding groove, the engaging member **210** is slidably connected in the sliding groove. The inner wall of the sliding groove forms the guide rail, to allow the engaging member **210** to slide along the inner wall of the sliding groove. Specifically, in some embodiments, the side of the lower wall surface of the pull-cover body **100** facing away from the first accommodation space is connected with a limiting plate **120**, the bottom of the pull-cover body **100** is recessed upward to form the first accommodation space, and the limiting plate **120** is arranged at the bottom of the first accommodation space. In this case, the upper surface of the limiting plate **120** forms the lower wall surface of the first accommodation space. In another embodiment, the side of the lower wall surface of the pull-cover body **100** facing away from the first accommodation space is connected with a limiting plate **120**, the bottom of the pull-cover body **100** is recessed toward the direction of the first accommodation space, to form the sliding groove between the limiting plate **120** and the pull-cover body **100**. The engaging member **210** slides in the sliding groove, the inner wall of the sliding groove and the limiting plate **120** form the guide rail, and the engaging member **210** moves along the sliding groove.

As shown in FIG. **23**, a second avoiding groove **121** is defined in the center of the limiting plate **120**. The second avoiding groove **121** is configured to avoid the handle portion **2500** of the cover body **2000** as shown in FIG. **16**.

In the embodiments, the first cam **420** rotates to cause the protruding portion **421** to abut against the pushing portion **212**, which in turns drives the engaging member **210** to move, and allows the engaging portion **211** to extend out of the pull-cover body **100** to engage with the cover body **2000**. As the first cam **420** continues to rotate to allow the recessed portion **422** to face the pushing portion **212**, the engaging member **210** needs to retract to allow the pushing part **212** to abut against the recessed portion **422**. In the embodiments, the snap-fit assembly **200** further includes a reset member **220**. Specifically, the reset member **220** has a certain elasticity, and the reset member **220** is connected to the engaging member **210**. When the protruding portion **421** drives the engaging member **210** to move, the reset member **220** is compressed and thus has a certain elastic force. As the first cam **420** continues to rotate to allow the recessed portion **422** to face the pushing portion **212**, the reset member **220** returns to its original state and releases the elastic force, thereby driving the engaging member **210** to move in the opposite direction. This action cooperates with the rotation of the first cam **420** to allow the pushing portion **212** to abut against the recessed portion **422**. The reset

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member 220 may be an elastic material, such as rubber, silicone, or a spring, which is not limited in the embodiments.

In a specific implementation process, the engaging member 210 is further provided with a fixing portion 215 extending upward along the axial direction of the second motor 410. One end of the reset member 220 is connected with the pull-cover body 100, and the other end of the reset member 220 is connected with the fixing portion 215. The reset member 220 applies an elastic resetting force to the engaging member 210, allowing it to switch from the locked state to the released state. Specifically, the inner wall of the first accommodation space of the pull-cover body 100 is provided with a mounting plate. The lower wall surface of the first accommodation space is defined with an elongated hole, and the extension direction of the elongated hole is consistent with the extension direction of the engaging member 210. The top surface of the engaging member 210 protrudes upward to form the fixing portion 215, and the fixing portion 215 is slidable in the elongated hole. One end of the reset member 220 is connected with the mounting plate, and the other end of the reset member 220 is connected with the fixing portion 215, such that the engaging member 210 compresses the reset member 220 during its movement from the released state to the locked state, thereby allowing the reset member 220 to have a restoring force to return to its original state. When the engaging member 210 is switched from the locked state to the released state, the protruding portion 421 of the first cam 420 gradually moves away from the pushing portion 212. In this case, the reset member 220, due to its restoring force, returns to its original state, thereby driving the engaging member 210 to move in the opposite direction. In this way, the pushing portion 212 of the engaging member 210 abuts against the recessed portion 422 of the first cam 420; that is, the engaging member 210 is in the released state. It can be understood that the end surface of the pushing portion 212 and the side surface of the protruding portion 421 are both arc-shaped, to ensure a smooth switch between the locked state and the released state of the engaging member 210.

In the embodiments, as shown in FIG. 23, the engaging portion 211 is provided with a first guide inclined surface 213 and a support surface 214 connected to each other. The first guide inclined surface 213 extends upward from the edge of the engaging member 210 to the support surface 214. When the pull-cover body 100 moves toward the first position, the first guide inclined surface 213 is configured to cooperate with the cover body 2000 to guide the cover body 2000 into the support surface 214.

It can be understood that, in practice, the placement position of the can is not precise, yet the pull-cover body 100 needs to cooperate with the cover body 2000 of the can to ensure that the engaging portion 211 of the engaging member 210 can be smoothly engaged in the groove of the cover body 2000. In the embodiments, the first guide inclined surface 213 is cooperated with the cover body 2000, which is conducive to the cooperation between the peripheral surface of the pull-cover body 100 and the top of the cover body 2000, making it easier for the engaging portion 211 to insert into the cover body 2000. Additionally, the support surface 214 abuts against the cover body 2000, to ensure that the pull-cover body 100 can smoothly pull the cover body 2000.

Further, the bottom end of the pull-cover body 100 is provided with a guide portion 110 for cooperating with the cover body 2000. The guide portion 110 is configured to move in the inner peripheral wall of the cover body 2000, to

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allow the pull-cover body 100 to move into or out of the cover body 2000. The guide portion 110 is defined with a first through hole 111. In the locked state, the engaging portion 211 can extend out of the first through hole 111 and engage with the groove 2400 of the cover body 2000. In a specific implementation process, the guide portion 110 is arranged at a bottom edge of the pull-cover body 100, and is arranged along the circumferential direction of the pull-cover body 100. In the released state, the engaging portion 211 is located in the first through hole 111, or, an end of the engaging portion 211 is flush with the guide portion 110, preventing the engaging portion 211 from affecting the entry of the pull-cover body 100 into the cover body 2000.

In a more specific implementation process, as shown in FIG. 4, the cover body 2000 includes a side wall portion 2100, a cover tray portion 2200, and a cover flange portion 2300. The cover tray portion 2200 is connected to the inner peripheral wall of the side wall portion 2100. The cover tray portion 2200 allows the cover body 2000, along an up-down direction of the cover body 2000, to be defined with a recessed space having an opening opened upward and a lower space having an opening opened downward. The cover flange portion 2300 is arranged in the recessed space. One end of the cover flange portion 2300 is connected with the side wall portion 2100, and the other end of the cover flange portion 2300 extends inward in a direction away from the cover body 2000. The cover flange portion 2300, the side wall portion 2100 and the cover tray portion 2200 together form the groove 2400. In the locked state, the guide portion 110 extends into the groove 2400 and is engaged with the cover flange portion 2300, such that when the pull-cover body 100 rises, it can pull the cover body 2000 to move.

Further, the guide portion 110 is provided with a second guide inclined surface 112. The second guide inclined surface 112 is inclined from the bottom of the pull-cover body 100 in a vertical upward direction toward the direction close to the press-can apparatus. When the pull-cover body 100 moves to the first position, the second guide inclined surface 112 is configured to cooperate with the cover body 2000, to move the pull-cover body 100 into the cover body 2000. Specifically, the second guide inclined surface 112 and the first guide inclined surface 213 are both located at the edge of the cover body 2000; besides, the second guide inclined surface 112 is connected to the outer side wall and the outer bottom wall of the pull-cover body 100 and is in contact with the cover body 2000, serving as a guide. This allows the pull-cover body 100 to smoothly enter and exit the cover body 2000. In the embodiments, the second guide inclined surface 112 is cooperatively arranged with the first guide inclined surface 213, and the two have the consistent inclination direction and inclination angle, to allow the bottom of the pull-cover body 100 to smoothly enter and exit the cover body 2000.

In some embodiments, the second driving assembly 400 further includes a second cam 430. The first cam 420 includes a cam shaft 423, and the second cam 430 is sleeved on the cam shaft 423. The second motor 410 drives the first cam 420 and the second cam 430 to move synchronously. The animal food dispensing device further includes a first detection assembly 130, which includes a first signal emitter 131 and a first signal receiver 132. The first signal emitter 131 is fixedly connected to the second cam 430, and the first signal receiver 132 is fixedly connected to the pull-cover body 100. In the locked state or/and the released state, the first signal emitter 131 is provided in correspondence with the first signal receiver 132.

In a specific implementation process, the second cam 430 is located above the first cam 420, and the cam shaft 423 of the first cam 420 extends upward along the axis direction, to allow the second cam 430 to be sleeved on the camshaft 423. Besides, a limiting surface is provided between the cam shaft 423 and the second cam 430, to prevent circumferential rotational displacement between the first cam 420 and the second cam 430 during rotation, thereby allowing the second motor 410 to drive the first cam 420 and the second cam 430 to rotate synchronously. In the embodiments, the first detection assembly 130 is configured to detect the action of the snap-fit assembly 200. Specifically, through the rotation of the second cam 430, the first signal emitter 131 triggers the first signal receiver 132, achieving detection on the action of the snap-fit assembly 200. Specifically, the first signal emitter 131 is fixedly connected with the second cam 430, and the second cam 430 moves synchronously with the first cam 420. When the first cam 420 drives the protruding portion 421 to allow the engaging member 210 in the locked state, the first signal emitter 131 fixedly connected with the second cam 430 is provided in correspondence with the first signal receiver 132; and in this case, the first detection assembly detects that the snap-fit assembly 200 is in the locked state. Or, when the first cam 420 drives the recessed portion 422 to allow the engaging member 210 in the released state, the first signal emitter 131 fixedly connected with the second cam 430 is provided in correspondence with the first signal receiver 132; and in this case, the first detection assembly detects that the snap-fit assembly 200 is in the released state.

In the embodiments, referring to FIG. 17, the first detection assembly 130 includes two first signal emitters 131. The first signal emitter 131 located above the protruding portion 421 is configured to detect whether the snap-fit assembly is in the locked state, and the first signal emitter 131 located above the recessed portion 422 is configured to detect whether the snap-fit assembly is in the released state.

In the embodiments, referring to FIG. 17, the first detection assembly 130 is mechanical switches. The first signal emitter 131 is a mechanical protrusion, and the first signal receiver 132 is a mechanical button. Specifically, the mechanical protrusion is arranged on the peripheral side wall of the second cam 430. In practice, the protrusion is arranged on the peripheral side wall of the second cam 430; and the mechanical button is arranged on the pull-cover body 100 and located on one side of the second cam 430. In this way, the rotation of the second cam 430 causes the mechanical protrusion to contact the mechanical button, which in turn triggers the first signal receiver 132 to generate a first detection signal. As needed, multiple mechanical protrusions may be spaced apart on the peripheral side wall of the second cam 430 along the circumferential direction. In this way, one full rotation of the second cam 430 can cause multiple contacts between the mechanical protrusions and the mechanical button, meeting the detection requirements for the snap-fit assembly 200. In the embodiments, the number of the mechanical protrusions is the same as the number of the protruding portions 421, and the angle of the extension protruding direction of the mechanical protrusions is consistent with that of the protruding portions 421.

Further, the first detection assembly 130 includes two first signal receivers 132. The number of the first signal emitters 131 is the same as the number of the engaging members 210 and is denoted as N. The first signal emitters 131 and the engaging members 210 are evenly spaced along the axis of the second motor 410 within a projection on a horizontal plane. The line connecting the axis of the second motor 410

to one of the first signal receivers 132 is L1, and the line connecting the axis of the second motor 410 to the other first signal receiver 132 is L2, wherein the central angle defined between L1 and L2 is α , and α and N satisfy the following equation: $\alpha=180/N$.

In the embodiments, N is 4 as an example, then, α is 45° , and the first signal emitters 131 are evenly spaced on the second cam 430. When the first signal emitter 131 triggers the first signal receiver 132 located above the protruding portion 421, the first cam 420 drives the engaging member 210 to extend to engage with the groove 2400 of the cover body 2000. In this case, the snap-fit assembly 200 is in the locked state, and the first driving assembly 300 can drive the pull-cover body 100 to rise and drive the cover body 2000 to rise synchronously. The second cam 430 continues to rotate by 45 degrees. The first signal emitter 131 triggers the first signal receiver 132 located above the recessed portion 422, and the engaging member 210 cooperates with the recessed portion 422 of the first cam 420. In this case, the snap-fit assembly 200 is in the released state, and the first driving assembly 300 can further drive the pull-cover body 100 away from the cover body 2000. During this process, the pull-cover body 100 moves from the first position to the second position to pull the cover body 2000 away from the can body 1000; then, it moves back from the second position to the first position to place the cover body 2000 back onto the can body 1000.

Further, the first driving assembly 300 includes a first motor 310 and a first transmission member which are in transmission connection with the pull-cover body 100. The first motor 310 drives the first transmission member to move vertically. The pressing member 600 is in the shape of a sleeve with an internal second accommodation space. One end of the pressing member 600 close to the can is open, and the other of the pressing member 600 is defined with a second through hole penetrating up and down. The pull-cover body 100 is arranged in the second accommodation space. The first transmission member passes through the second through hole 611 to drive the pull-cover body 100 to move vertically.

The pull-cover body 100 is located in the second accommodation space of the pressing member 600 and moves in the second accommodation space. It can be understood that the bottom end of the pressing member 600 is open, which facilitates the connection between the pull-cover body 100 and the cover body 2000 to open the cover body 2000. The pressing member 600 is sleeved outside the pull-cover body 100, and the bottom surface of the pressing member 600 abuts against the support portion of the can body 1000. It can be understood that, the outer wall of the support portion is located outside the pressing member 600 within a projection on a horizontal plane, to ensure effective abutment between the pressing member 600 and the support portion, thereby effectively limiting the vertical movement of the can body 1000.

As shown in FIG. 21, the top of the pressing member 600 is opened with the second through hole 611. In some embodiments, the pressing member 600 is provided with an integrally formed top, the second through hole 611 is arranged on the top. The first motor 310 is mounted on the top of the pressing member 600 and located on the surface facing away from the second accommodation space. In another embodiment, the animal food dispensing device further includes a fixing member 610. The pressing member 600 and the pull-cover body 100 are fixedly connected to the fixing member 610. The fixing member 610 is connected to the top of the pressing member 600 by means of screw

connection or engagement connection, such that the second accommodation space is closed at its top. The first motor 310 is mounted on the side of the fixing member 610 facing away from the fixing member 610. The first motor 310 drives the pull-cover body 100 to move between the first position and the second position by the first transmission member. The first transmission member may be a gear transmission mechanism, or a conveyor belt mechanism, or the like, which is not limited herein, as long as it enables the pull-cover body 100 to move up and down under the driving action of the first motor 310. In the embodiments, the fixing member 610 is connected to both the pressing member 600 and the pull-cover body 100, such that the movement of the pressing member 600 drives the pull-cover body 100 to move. Specifically, according to the animal food dispensing device in the embodiments, in the action of separating the can body 1000 from the cover body 2000, the pressing member 600 moves downward to abut against the support portion of the can body 1000. The pressing member 600 drives the pull-cover body 100 to move downward, and then the pull-cover body 100 moves downward to reach the first position. After the snap-fit assembly 200 is engaged with the cover body, the pull-cover body 100 moves upward to achieve the separation of the can body 1000 and the cover body 2000.

In the embodiments, the bottom of the pressing member 600 is provided with a flanging 620 extending outward along the circumferential direction. The flanging 620 is configured to abut against the support portion of the can body 1000, to increase the contact area between the pressing member 600 and the support portion. In the embodiments, to ensure the effectiveness of the abutment, the top surface of the support portion is flat, which is convenient for the flanging 620 to abut against the support portion. Specifically, the bottom end of the pressing member 600 bends away from the axis to form the flanging 620. The pressing member 600 surrounds the pull-cover body 100 and the cover body 2000 inside the second accommodation space. The flanging 620 abuts against the support portion to limit the upward vertical displacement of the can body 1000.

Further, the fixing member 610 is opened with a second through hole 611. The first transmission member includes a first gear 320 and a first rack 330. The first motor 310 is mounted on the fixing member 610. An output shaft of the first motor 310 is in transmission connection with the first gear 320. The first rack 330 has a first surface and a second surface facing the first surface. The first surface is meshed with the first gear 320 for transmission. The surface of the fixing member 610 facing the first motor 310 is provided with a limiting portion 612, and the limiting portion 612 is arranged close to the second surface to limit the first rack 330 to move in a direction away from the first gear 320. The first rack 330 passes through the second through hole 611 to drive the pull-cover body 100 to move vertically.

In a specific implementation process, the first rack 330 extends in the vertical direction. One end of the first rack 330 is connected to the top of the pull-cover body 100, and the other end of the first rack 330 passes through the second through hole 611, and the first rack 330 is allowed to slide along the second through hole 611. The first motor 310 drives the first rack 330 to move in the vertical direction by the first gear 320, to allow the pull-cover body 100 to move between the first position and the second position. The fixing member 610 is connected with a limiting portion 612, which is in contact with the second surface of the first rack 330, to prevent horizontal displacement of the first rack 330. On the one hand, this can prevent the pull-cover body 100 from

swaying in the horizontal direction, enhancing the movement stability of the first rack 330 and the pull-cover body 100. On the other hand, it allows the first rack 330 to be always meshed with the first gear 320, ensuring the effectiveness of the first motor 310 driving the movement of the first rack 330. It can be understood that the first surface of the first rack 330 is provided with gear teeth for meshing with the first gear 320, and the second surface of the first rack 330 is flat to facilitate cooperation with the limiting portion 612.

In some embodiments, the limiting portion 612 includes a third roller 613. The third roller 613 is in rolling operation with the first rack 330 to reduce friction. Specifically, the peripheral side wall of the third roller 613 is in contact with the second surface of the first rack 330, such that the first rack 330 is in rolling operation with the third roller 613, thereby reducing friction. In a specific implementation process, the third roller 613 is fixed on the fixing member 610 and rotatable relative to the fixing member 610. In another embodiment, the limiting portion 612 includes a rolling ball that is in contact with the first rack 330, which can further reduce friction.

In some embodiments, the animal food dispensing device further includes a second detection assembly 630, which includes a second signal emitter 631 and a second signal receiver 632. The second signal emitter 631 is fixedly connected to the first rack 330, and the second signal receiver 632 is fixedly connected to the fixing member 610. The second detection assembly 630 is configured to cooperate with the first driving assembly 300, to control the movement of the pull-cover body 100.

The second detection assembly 630 may be a photoelectric sensor, a mechanical switch, or the like, which is not limited herein, as long as it ensures smooth triggering. In a specific implementation process, the second detection assembly 630 is a mechanical switch, such as a limit switch. Specifically, the second signal emitter 631 is a mechanical protrusion, and the second signal receiver 632 is a mechanical button. At the first position, the second signal emitter 631 abuts against the second signal receiver 632. When the first rack 330 moves to a position where the second signal emitter 631 triggers the second signal receiver 632, the second signal receiver 632 is triggered to generate a second detection signal. In the embodiments, the second detection signal is the signal indicating that the pull-cover body 100 has reached the first position.

In some embodiments, there are two second signal emitters 631 and two second signal receivers 632. One second signal emitter 631 is arranged on the first rack 330 and located in the space above the fixing member 610, and one second signal receiver 632 is mounted on the fixing member 610 and located on the side of the fixing member 610 facing away from the second accommodation space; and in this case, the second signal emitter 631 is configured to trigger the second signal receiver 632. The other second signal emitter 631 is arranged on the first rack 330 and located in the space below the fixing member 610, or the other second signal emitter 631 is arranged on the top surface of the pull-cover body 100, and the other second signal receiver 632 is arranged on the fixing member 610 and located on the side of the fixing member 610 facing the second accommodation space; and in this case, the second signal emitter 631 is configured to trigger the second signal receiver 632. In another embodiment, one second signal emitter 631 is mounted on the first rack 330, and two second signal receivers 632 with a vertical spacing distance therebetween are arranged on the fixing member 610. When the pull-cover

body **100** is at the first position, the second signal emitter **631** triggers one second signal receiver **632**; and when the pull-cover body **100** is at the second position, the second signal emitter **631** triggers the other second signal receiver **632**.

Based on the aforementioned, when the pull-cover body **100** moves to the first position, the second signal emitter **631** is provided in correspondence with the second signal receiver **632** to generate a detection signal. When the pull-cover body **100** moves to the second position, the other second signal emitter **631** is provided in correspondence with the other second signal receiver **632** to generate a detection signal.

In some embodiments, the press-can apparatus includes a pressing member **600** and a third driving assembly **500**, and an outer peripheral wall of the can body **1000** is provided with a support portion. The third driving assembly **500** is in transmission connection with the pressing member **600**, to allow the pressing member **600** to have a third position and a fourth position in the vertical direction. At the third position, the pressing member **600** abuts against the support portion to limit vertical movement of the can body **1000**. At the fourth position, the pressing member **600** is separated from the support portion. The third driving assembly **500** drives the pressing member **600** to move between the third position and the fourth position.

In a specific implementation process, the support portion is fixed on the outer peripheral wall of the can body **1000** and protrudes from the outer peripheral wall, facilitating the pressing member **600** to move from top to bottom and abuts against the top surface of the support portion. Specifically, the bottom of the pressing member **600** abuts against the top surface of the support portion. The third driving assembly **500** drives the pressing member **600** to move up and down. The third position is located below the fourth position. The third driving assembly **500** drives the pressing member **600** to move from the fourth position to the third position, to allow the pressing member **600** to move from the direction away from the support portion toward the direction close to the support portion, thereby abutting against the support portion. This process serves as a fifth stroke of the pressing member **600**. The third driving assembly **500** drives the pressing member **600** to move from the third position to the fourth position, to allow the pressing member **600** to move from the support portion in a direction away from the support portion. This process serves as a sixth stroke of the pressing member **600**. The third driving assembly may be a motor-driven structure, or may be a cylinder lifting mechanism, which is not limited herein.

Further, the third driving assembly **500** includes a third motor **510** and a second transmission member. The second transmission member includes a second gear **520** and a second rack **530**. An output shaft of the third motor **510** is in transmission connection with the second gear **520**, the pressing member **600** is fixedly connected with the second rack **530**, and the second gear **520** is meshed with the second rack **530** to drive the pressing member **600** to move in the vertical direction.

In the embodiments, the third motor **510** drives the pressing member **600** to move between the third position and the fourth position by the second transmission member. Specifically, the second rack **530** extends vertically and is fixedly installed on the outer side wall of the pressing member **600**. The third motor **510** drives the second gear **520** to rotate, such that the second rack **530** moves in the vertical

direction, thereby realizing the movement of the pressing member **600** between the third position and the fourth position.

In some embodiments, the animal food dispensing device further includes a housing **900** in the shape of a sleeve with an internal third accommodation space. The third motor **510** is mounted on the housing **900**, and the press-can apparatus and the pull-cover apparatus are arranged in the third accommodation space. In a specific implementation process, the housing **900** is sleeved outside the pressing member **600** accommodated in the third accommodation space, and the third motor **510** is mounted on the housing **900**. The housing **900** is defined with a first avoiding groove **910** to avoid the second rack **530**, allowing the second rack **530** to slide in the first avoiding groove **910**.

It can be understood that the pull-cover body **100** is located in the second accommodation space of the pressing member **600**, and the pull-cover body **100** moves up and down in the second accommodation space. In this way, there is a certain gap between the outer wall of the pull-cover body **100** and the inner wall of the second accommodation space, which may cause the pull-cover body **100** to sway during its vertical movement and fail to enter the cover body **2000**. This could result in that the engaging member **210** is unable to engage with the cover body **2000**, thereby failing to open the can. In the embodiments, the animal food dispensing device further includes a first walking assembly **700**. The first walking assembly **700** includes a first roller **710**, which is assembled to the pull-cover body **100**. The first roller **710** may be in rolling cooperation with the pressing member **600** to reduce friction generated when the pull-cover body **100** and the pressing member **600** move relative to each other.

Referring to FIG. **21** and FIG. **23**, in the embodiments, the first walking assembly **700** is arranged on the pull-cover body **100** and contacts the pressing member **600** to limit the horizontal displacement of the pull-cover body **100**, thereby providing a good limiting and guiding effect on the pull-cover body **100**. Specifically, the first walking assembly **700** further includes a first mounting base **720**, which is fixed on the side wall of the pull-cover body **100** by a screw. The first roller **710** is mounted in the first mounting base **720**, and the outer peripheral wall of the first roller **710** is in contact with the inner wall of the pressing member **600**. During the process of the pull-cover body **100** moving up and down, the first roller **710** rolls along the pressing member **600**, which reduces the friction between the pull-cover body **100** and the pressing member **600**, and also limits the pull-cover body **100**, thereby enhancing the stability of the up and down movement of the pull-cover body **100**. It can be understood that, according to actual needs, the first mounting base **720** may be provided with multiple first rollers **710** to further improve the stability of the pull-cover body **100**. In addition, multiple first walking assemblies **700** are spaced apart along the circumferential direction of the pull-cover body **100** to ensure the balance of the pull-cover body **100**. The first roller **710** may be made of a soft material such as rubber or silicone, which reduces wear and noise, and also helps to save costs.

The animal food dispensing device further includes a second walking assembly **800**. The second walking assembly **800** includes a second roller **810**, which is assembled on the housing **900**. The second roller **810** may be in rolling operation with the pressing member **600** to reduce the friction generated when the pressing member **600** moves relative to the housing **900**.

Referring to the aforementioned, in a specific implementation process, the pressing member **600** moves in the third

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accommodation space of the housing 900. In this way, there is also a certain gap between the outer wall of the pressing member 600 and the inner wall of the third accommodation space, which may cause the pressing member 600 to sway during its vertical movement and fail to accurately abut against the support portion of the can body 1000. This could result in that the pressing member 600 is unable to limit the can body 1000, thereby failing to open the can. In the embodiments, the second walking assembly 800 is arranged between the pressing member 600 and the housing 900 to limit the horizontal displacement of the pressing member 600, ensuring smooth opening of the can body 1000.

Specifically, the second walking assembly 800 further includes a second mounting base 820, which is fixed on the side wall of the pressing member 600 by a screw. The second roller 810 is mounted in the second mounting base 820, and the outer peripheral wall of the second roller 810 is in contact with the inner wall of the housing 900. During the process of the pressing member 600 moving up and down, the second roller 810 rolls along the pressing member 600, which reduces the friction between the pressing member 600 and the housing 900, and also limits the pressing member 600, thereby enhancing the stability of the up and down movement of the pressing member 600. It can be understood that, according to actual needs, the second mounting base 820 may be provided with multiple second rollers 810 to further improve the stability of the pressing member 600. In addition, multiple second walking assemblies 800 are spaced apart along the circumferential direction of the pressing member 600 to ensure the balance of the pressing member 600. The second roller 810 may be made of a soft material such as rubber or silicone, which reduces wear and noise, and also helps to save costs.

In some embodiments, the animal food dispensing device further includes a third detection assembly 920. The third detection assembly 920 includes a third signal emitter 921 and two third signal receivers 922. The third signal emitter 921 is fixedly connected to the press-can apparatus, and the third signal receivers 922 are fixedly connected to the housing 900. At the third position, one of the two third signal receivers 922 is provided in correspondence with the third signal receiver 922; and at the fourth position, the other of the two third signal receivers 922 is provided in correspondence with the third signal receiver 922.

The third detection assembly 920 cooperates with the third driving assembly 500 to control the movement of the pressing member 600. Specifically, the third signal emitter 921 may be arranged on the second rack 530, and moves with the second rack 530; or, the second rack 530 serves as the third signal emitter 921. The two third signal receivers 922 are arranged on the housing 900, and there is a certain distance, namely the movement distance of the pressing member 600, between the two third signal receivers 922. In the embodiments, the pressing member 600 reaches the fourth position, the rack is located between the two third signal receivers 922, and the pressing member 600 moves upward, so that when the top end of the rack 922 is sensed by one third signal receiver, this third signal receiver 922 is triggered to generate a third detection signal to determine the position condition of the pressing member 600. Similarly, the pressing member 600 is located at the third position, when the bottom end of the rack is sensed by the other third signal receiver 922, this third signal receiver 922 is triggered, the third motor 510 stops, and a third detection signal is generated to determine the position condition of the pressing member 600.

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In a specific implementation process, the third detection assembly 920 is a photoelectric sensor, the third signal emitter 921 is a photoelectric emitter, and the third signal receiver 922 is a photoelectric receiver; or, the third detection assembly 920 is a magnetic field sensor, the third signal emitter 921 is a magnetic signal emitter, and the third signal receiver 922 is a magnetic signal receiver; or, the third detection assembly 920 is a mechanical switch, the third signal emitter 921 is a mechanical protrusion, and the third signal receiver 922 is a mechanical button, which is not detailed herein.

The foregoing are only optional embodiments of the present disclosure, and are not intended to limit the patent scope of the present disclosure. Under the inventive concept of the present disclosure, any equivalent structural transformation made by using the contents of the specification and drawings of the present disclosure, or directly/indirectly applied in other related technical fields, are included in the patent protection scope of the present disclosure.

What is claimed is:

1. A food container, comprising:

a can body comprising a container body, the container body comprising a container bowl portion, the container bowl portion being defined with an accommodation chamber for holding food and an opening opened upward and communicated with the accommodation chamber; and

a cover body, the cover body comprising a side wall portion, a cover tray portion, and a cover flange portion; the cover tray portion being connected to an inner peripheral wall of the side wall portion, the cover tray portion allowing the cover body, along an up-down direction of the cover body, to be defined with a recessed space having an opening opened upward and a lower space having an opening opened downward, the cover flange portion being arranged in the recessed space, an end of the cover flange portion being connected to the side wall portion, the other end of the cover flange portion extending inward in a direction away from the side wall portion; and the cover flange portion, the side wall portion, and the cover tray portion defining a groove;

wherein the cover body is assembled to the can body, and the container body is snap-fitted with the side wall portion;

wherein the food container further comprises:

a sealing membrane for sealing the accommodation chamber;

wherein a portion of the sealing membrane is connected to the cover tray portion, and at least part of the other portion of the sealing membrane is connected to the container bowl portion.

2. The food container according to claim 1, wherein the sealing membrane comprises:

a sealing membrane body; and

a connection lug portion extending outward from an outer peripheral edge of the sealing membrane body;

wherein the sealing membrane body is connected to an upper surface of the container bowl portion, and the connection lug portion is folded upward and inward to be connected to a lower surface of the cover tray portion.

3. The food container according to claim 2, wherein

the connection lug portion comprises a first adhesion section and a second adhesion section connected to the first adhesion section, and the first adhesion section is

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closer to an outer peripheral edge of the sealing membrane body than the second adhesion section; and the connection lug portion is one of two connection lug portions, the first adhesion section of one of the two connection lug portions is connected to the lower surface of the cover tray portion, and the second adhesion section of the other one of the two connection lug portions is connected to the lower surface of the cover tray portion.

4. The food container according to claim 2, wherein the connection lug portion is one of a plurality of connection lug portions, and the plurality of connection lug portions are symmetrically arranged about a central axis of the sealing membrane.

5. The food container according to claim 1, wherein the sealing membrane has a first surface and a second surface facing the first surface, the first surface of the sealing membrane is connected to an upper surface of the container bowl portion, and the second surface of the sealing membrane is connected to a lower surface of the cover tray portion;

or,

the cover tray portion is defined with a first through hole, the sealing membrane comprises a sealing membrane body and a connection lug portion extending outward from an outer peripheral edge of the sealing membrane body, the sealing membrane body is connected to an upper surface of the container bowl portion, the connection lug portion passes through the first through hole into the recessed space, and the connection lug portion is connected to an upper surface of the cover tray portion.

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6. The food container according to claim 1, wherein the cover body is further provided with a handle portion arranged in the recessed space, the handle portion is connected to the cover tray portion, and an upper surface of the handle portion is lower than the cover flange portion in an axial direction of a central axis of the cover body.

7. The food container according to claim 1, wherein the cover body is further provided with a first flange portion arranged in the lower space, an end of the first flange portion is connected to the side wall portion, and the other end of the first flange portion extends inward in a direction away from the side wall portion; and

the container body is further provided with a second flange portion, an end of the second flange portion is connected to the container bowl portion, and the other end of the second flange portion extends outward in a direction away from the container body;

wherein the cover body is assembled to the can body, and the first flange portion is snap-fitted with the second flange portion.

8. The food container according to claim 7, wherein the second flange portion forms an upper peripheral edge of the container body; and

the cover body is assembled to the can body, and an inner edge of the cover flange portion fails to protrude beyond an inner surface of the second flange portion within a projection of the food container on a horizontal plane.

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